Bounding Outcomes of Repeated Games with Private Monitoring

Avner A. Kreps*

August 17, 2026

Abstract

This paper studies bounds on outcomes in repeated games with private monitoring. The main result places a necessary condition on equilibrium outcomes that relates the short-run gain from a deviation to the discount factor, the statistical detectability of a deviation in rivals' signals, and the on-path payoff variance. This result is sharper than existing bounds when players are patient and their signals contain private information. The bound also accommodates settings where a portion of signals are observed on a delay. I use these results to study how information exchanges affect the potential for firms to collude, a leading application of repeated games with private monitoring.

JEL: C73, D82, L41

Keywords: Repeated games, private monitoring, communication game

1 Introduction

It is well-established in the theory of repeated games that imperfect monitoring can inhibit efficiency. However, computing the exact set of equilibria in such games is notoriously intractable, and hence exactly characterizing the way in which the set of equilibrium payoffs in such games varies with the monitoring structure is known to be difficult. This has motivated a nascent literature that has derived bounds on equilibria in repeated games with private monitoring structures (Sugaya and Wolitzky, 2018; 2023; Awaya and Krishna, 2016; 2019). These bounds take the form of necessary conditions that equilibria must satisfy. To use these bounds to form predictions about how monitoring changes the set of equilibria, it is desirable

*Northwestern University, Department of Economics. Email: avner@u.northwestern.edu. I thank Vivek Bhattacharya, Jeff Ely, Gastón Illanes, Harry Pei, and Rob Porter for helpful comments.

that these necessary conditions are as close as possible to sufficient conditions, making the bounds they imply as sharp as possible.

This paper proposes a new bound on equilibrium outcomes in repeated games with private monitoring structures that is sharper than existing bounds under relatively general conditions. The main result can be described heuristically as follows. Suppose that after each period, all players communicate about their signals. In one period, a player decides to deviate from her prescribed equilibrium action and lie about the signal she receives in an attempt to "cover up" the deviation. If such a deviation is unprofitable—as it must be in equilibrium—then the expected short-run increase in payoffs from this deviation is bounded by

$$\sqrt{\frac{\delta}{1-\delta}(\text{on-path payoff variance})(\text{detectability of deviation using reported signals})}$$

where δ is the discount factor. This result most directly relates to Sugaya and Wolitzky (2023) (henceforth SW23), who present a similar bound in a setting where a mediator observes the true signals, and hence players cannot lie about them.[1] Because my framework allows players to misreport their signal, this result is more general than SW23's, and in particular is sharper in settings where players' signals contain private information about their action, which is arguably the setting of interest in private monitoring games.

This result facilitates comparison to the bound of another related paper, Awaya and Krishna (2019) (henceforth AK19). AK19 prove a bound based on a fixed deviation to an alternative strategy, also relating a measure of monitoring precision to patience and the range of payoffs. While AK19's and SW23's bounds have several similarities—both use statistical measures of the difference between distributions to measure monitoring precision that are of similar magnitudes—there are two key differences. First, AK19's bound is of order $\delta/(1-\delta)$, as it bounds incentives history-by-history, while SW23's bound (like the one in this paper) is of order $\sqrt{\delta/(1-\delta)}$, as it bounds incentives on average across histories. This makes SW23's bound relatively tighter for more patient players. However, AK19's bound, by using an information-free deviation to a single action, depends on the monitoring efficacy of rivals' signals, while SW23's bound, which is based on a one-period deviation, depends on the monitoring efficacy of the entire vector of signals, which is looser in cases when players' signals carry private information about their action. This is exactly the dimension along which this paper's bound improves on SW23. As such, while SW23's bound and AK19's bound are nonnested, I derive conditions under which this paper's bound is uniformly sharper than AK19's, finding that if players' conditional signal distributions under a deviation

[1] SW23 mention the possibility of this strengthening in their paper (p. 1735); I formally state it and analyze the existence and magnitude of the tightening that it provides.

Blackwell-dominate those under on-path strategies, then my bound is sharper than AK19's in the patient limit.

I also extend the bounds to accommodate cases in which signals are observed on a delay. Prior papers have studied the impact of delaying signals in repeated games with perfect or public monitoring structures (Abreu et al., 1991; Fudenberg et al., 2014; Igami and Sugaya, 2022), but their impact on outcomes in private monitoring games has yet to be studied. If part of the signal is observed immediately and part of the signal is observed on a delay of L periods, I find that all three bounds (AK19's, SW23's, and this paper's) go through verbatim with detectability replaced by the *effective detectability*, defined as

$$\begin{aligned}\text{effective detectability} = (1-\delta^L)\cdot \text{detectability of immediate signal} +\\ \delta^L \cdot \text{detectability of full signal.}\end{aligned}$$

In other words, monitoring capabilities "discount" at the same rate as payoffs. This expression has a suitable generalization for cases in which signals are observed over more than two periods with arbitrarily heterogeneous timing. The intuition is that deviation only causes changes to future payoffs to the extent that players can use their signals to detect them. If some signals are not observed for L periods, then they can only impact future payoffs after L periods, at which point payoffs are discounted by δ^L.

Using these results, I study a leading application of repeated games with private monitoring: the effects of information sharing on tacit collusion. When firms exchange information with one another, antitrust enforcers are typically concerned that the additional monitoring capabilities provided by that information facilitate collusion, and have historically demanded the degradation of these monitoring structures as remedies (Harrington, 2026).[2] I demonstrate that in a symmetric log-linear demand setting, three different monitoring structures of interest—no observation of rivals' sales, full observation of rivals' sales, and public observation of a demand signal—have effects on the bound that boil down to a single constant with a closed form. Changes to this constant across monitoring structures can directly be traded off with the discount factor; thus, one can directly state the difference between monitoring structures in terms of a difference in discount factors that would generate the same change in the set of equilibria that satisfy the bound. Comparative statics suggest that in industries with demand that is more correlated across firms, public demand information can effectively facilitate collusion, while verified sales information is needed to effectuate collusion in settings with more uncorrelated demand. These results provide new testable implications for the structure

[2]For example, the settlement of *US et al. v. Agri Stats, Inc.*, a case about an information exchange in the poultry and meat processing industry, requires that firms only share aggregate sales information as well as delay any exchanged information so that it is at least 45 days old (US et al. v. Agri Stats, Inc., 2026).

of real-world information exchanges.

Beyond the clear relations to AK19 and SW23, this paper relates more broadly to the literature on repeated games with private monitoring structures (Kandori and Matsushima, 1998; Kandori, 2002; Mailath and Samuelson, 2006). Sugaya (2022) proves a folk theorem for repeated games with private monitoring and sufficiently broad signal structures, although the equilibrium strategies are not guaranteed to be optimal in games with finitely patient players. Earlier papers bounding incentives in repeated games include Fudenberg et al. (1998), Al-Najjar and Smorodinsky (2000), Al-Najjar and Smorodinsky (2001), Awaya and Krishna (2016), and Sugaya and Wolitzky (2018). Specific to information exchanges and collusion, Awaya and Krishna (2020) study conditions under which exchanges of aggregate sales information can result in strictly higher equilibrium payoffs, while Kreps (2026) uses the results of this paper to bound the effects of counterfactual exchanges within the environment of a particular real-world information exchange. This paper contributes a framework for studying possible outcomes in repeated games that, while falling short of exactly classifying the set of equilibria, is sharper than any previously derived under reasonably general conditions.

2 Preliminaries

2.1 Environment

A *stage game* G is characterized by a tuple (I, A, u), where $I = \{1, \ldots, N\}$ is the set of players, $A = \prod_i A_i$ is a finite set of action profiles, and $u : A \to \mathbb{R}^N$ gives payoffs for each player as a function of the action profile. Payoffs are bounded: for all i and $a \in A$, $|u_i(a)| \leq \bar{u} < \infty$. α denotes a distribution over A; let the mean and variance of i's payoff given such a distribution be denoted by $u_i(\alpha) = \mathbb{E}_{a\sim\alpha}[u_i(a)]$ and $V_i(\alpha) = \mathrm{Var}_{a\sim\alpha}(u_i(a))$.

A *monitoring structure* $M = (Y, p, \mathcal{L})$ is defined by a set of possible signal realizations for all players $Y = \prod_i Y_i$, a family of conditional distributions $p(\cdot \mid a) \in \Delta(Y)$ for all $a \in A$, and a *lag structure* $\mathcal{L}$ that factors each player's signal space into $L + 1$ components, $Y_i = \prod_{\ell=0}^{L} Y_i^\ell$. The idea is that after players play the stage game with action profile a, player i receives signal $y_i^\ell \sim p_i^\ell(\cdot \mid a)$ ℓ periods later, where p_i^ℓ is the marginal over $y_i^\ell \in Y_i^\ell$ from p. Write $Y_\ell = \prod_i Y_i^\ell$, so that $Y = \prod_{\ell=0}^{L} Y_\ell$. I assume a common support; that is, there exists $\bar{Y} \subseteq Y$ such that for all y and a, $p(y \mid a) > 0$ if and only if $y \in \bar{Y}$. Two special cases are *undelayed monitoring structures*, in which $L = 0$, and *two-part signals with lag L*, in which Y_ℓ is degenerate for $\ell \notin \{0, L\}$.

A *repeated game* $\Gamma = (G, M, \delta)$ is defined by a stage game G, a monitoring structure M, and a common discount factor $\delta \in (0, 1)$. In each period $t = 1, 2, \ldots$ of the repeated game, each player i simultaneously selects an action $a_{i,t} \in A_i$. Player i then observes signals $y_{i,t} = (y^{\ell}_{i,t-\ell})^{\min\{t-1,L\}}_{\ell=0}$, where $(y^{\ell}_t)^L_{\ell=0}$ is drawn from $p(\cdot \mid a_t)$.[3] Thus, a history for player i takes the form $h^t_i = (a_{i,t'}, y_{i,t'})^{t-1}_{t'=1}$, and a strategy σ_i maps a history h^t_i into a distribution over $a_{i,t}$. Finally, a given strategy profile σ maps to an *outcome* μ, which is a distribution over infinite paths of actions and signals. Players attempt to maximize discounted expected payoffs.

Suppose a given outcome μ generates a distribution over actions α^{μ}_t in each period. The *occupation measure* over these distributions is the discounted average over these distributions; that is, for all $a \in A$,

$$\alpha^{\mu}(a) = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1}\alpha^{\mu}_t(a).$$

The occupation measure is a sufficient statistic for players' payoffs, as

$$(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}u_i(\alpha^{\mu}_t) = \sum_{a\in A}(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\alpha^{\mu}_t(a)u_i(a) = \sum_{a}\alpha^{\mu}(a)u_i(a) = u_i(\alpha^{\mu}).$$

The communication game. I consider the *communication game* Γ^C derived from a repeated game Γ. In the communication game, there exists a mediator that receives messages from each player about its actions and signals, and recommends actions to each player using this information. The timing of any given period t of the communication game is as follows:

1. The mediator sends recommendations $r_t = (r_{i,t})^N_{i=1}$ to each player. Player i observes only its own recommendation $r_{i,t}$.
2. Players simultaneously take actions $a_t = (a_{i,t})^N_{i=1}$.
3. Players observe signals $\left((y^{\ell}_{i,t-\ell})^{\min\{t-1,L\}}_{\ell=0}\right)^N_{i=1}$.
4. Players simultaneously send messages $m_t = (m_{i,t})^N_{i=1}$ to the mediator.

Player i's history when it is selecting an action takes the form $h^t_i = (r_{i,t'}, a_{i,t'}, y_{i,t'})^{t-1}_{t'=1}$. A (behavior) strategy for player i is a mapping from all such histories to tuples $(\alpha_i, m_i) \in \Delta(A_i) \times \mathcal{M}_i$, where α_i is a distribution over actions, m_i specifies a contingent message given a period-t action and observed signals, and $\mathcal{M}_i$ denotes the set of such contingent messages. Meanwhile, a history for the mediator when it is recommending an action in period t takes the form $h^t_0 = (r_{t'}, m_{t'})^{t-1}_{t'=1}$, and a strategy for the mediator maps such histories into recommendations for all players.

The goal is to find a necessary condition that any Nash equilibrium of Γ must satisfy. In order to do so, I analyze the communication game; since the set of Nash equilibria of Γ is a subset of those of Γ^C (Forges, 1986), a necessary condition for an equilibrium of the

[3]Note that y^{ℓ}_t is *generated* by actions in period t but is not *observed* until period $t+\ell$.

communication game is a fortiori a necessary condition for the original repeated game. By standard arguments (Myerson, 1986), any Nash equilibrium outcome is attainable as an obedient and truthful equilibrium; that is, one in which messages are reports of past signals that are truthful on-path and recommendations are actions that are followed on-path.[4] I therefore restrict attention to signal-valued messages and action-valued recommendations from this point forward.

On the other hand, SW23 focus on the blind game Γ^B. In the blind game, players remember their past actions but do not observe any signals, while the mediator does not observe actions but observes all past signals from all players. This makes signals verifiable—the mediator observes all signals directly and correctly, and players (who have no inherent informational advantage over the mediator beyond knowing their own actions) cannot use private information of their own signal realization to deviate. Meanwhile, in the communication game, players can give any information to the mediator about their actions and signals. This implies that the blind game supports a larger set of equilibria than the communication game.

2.2 Manipulations

An *action manipulation* for player i is a mapping $s_i : A_i \to \Delta(A_i)$. The idea is that when recommended action a_i, the player instead plays action $\widetilde{a}_i$ with probability $s_i(\widetilde{a}_i \mid a_i)$. The (expected flow) *gain* from an action manipulation s_i given an action distribution α is

$$g_i(s_i, \alpha) = \sum_a \sum_{\widetilde{a}_i} \alpha(a) s_i(\widetilde{a}_i \mid a_i) \left(u_i(\widetilde{a}_i, a_{-i}) - u_i(a)\right). \tag{1}$$

α is a *static ε-correlated equilibrium* action distribution if $g_i(s_i, \alpha) \leq \varepsilon$ for all i and s_i.

A *message manipulation* for player i is a set of mappings $q_i = (q_i^\ell)_{\ell=0}^L$, with $q_i^\ell : A_i^2 \times \prod_{\ell'=0}^{\ell} Y_i^{\ell'} \times \prod_{\ell'=0}^{\ell-1} Y_i^{\ell'} \to \Delta(Y_i^\ell)$ mapping from a recommended action a_i, an actual action $\widetilde{a}_i$, a set of prior signal realizations $(y_i^{\ell'})_{\ell'=0}^{\ell}$, and a set of prior reports $(\widetilde{y}_i^{\ell'})_{\ell'=0}^{\ell-1}$ to a probability distribution over reported signals. $q_i^\ell(\widetilde{y}_i^\ell \mid a_i, \widetilde{a}_i, y_i^0, \ldots, y_i^\ell, \widetilde{y}_i^0, \ldots, \widetilde{y}_i^{\ell-1})$ is the probability that player i reports signal $\widetilde{y}_i^\ell$ to the mediator ℓ periods after receiving action recommendation a_i, playing action $\widetilde{a}_i$, receiving signals $y_i^0, \ldots, y_i^\ell$, and having reported $\widetilde{y}_i^0, \ldots, \widetilde{y}_i^{\ell-1}$. Note that a message manipulation is specific to signals generated by actions in a particular period, not those observed in a particular period. I refer to a pair (s_i, q_i) as a *manipulation* for player i.

The analysis will consider the one-shot deviation of player i manipulating its action in one period and all messages of signals relating to that period according to (s_i, q_i), while

[4]The fact that recommendations are followed on-path means that the mediator effectively knows the actions, meaning players do not need to include actions in their messages.

being obedient and truthful in all other periods.[5] If this deviation is applied in period t, the deviator will send a period-t message that is in part truthful and in part not; it will truthfully report $y^{\ell}_{i,t-\ell}$ for $\ell > 0$ but manipulate $y^0_{i,t}$. One might wonder whether manipulating all signals at once would be a more effective deviation. However, note that due to the imperfect monitoring structure and full support assumption, rivals cannot detect deviation off-path, meaning players must be dissuaded from deviation using variation in on-path payoffs; there is no analogy to a grim-trigger punishment. Because of this, player i can attain a greater benefit from manipulating its reported signals if it also manipulates the actions generating those signals in a beneficial way. This provides a heuristic basis for why I consider a deviation in which player i only manipulates messages corresponding to a manipulated action, although there is no formal guarantee that this is the binding deviation, hence why the main result provides a necessary but perhaps insufficient condition.

2.3 Detectability

Because reports of signals are truthful on path, how a given manipulation affects the distribution of reported signals, not actual signals, is what matters for detectability. Suppose the mediator recommends action profile a, which player i manipulates according to (s_i, q_i), and action profile $\widetilde{a} = (\widetilde{a}_i, a_{-i})$ is played. Because players $-i$ report truthfully, their reports coincide with their true signals, and I write y_{-i} for both. The conditional probability of the mediator receiving a reported signal profile $\widetilde{y} = (\widetilde{y}_i, y_{-i})$ is

$$p(\widetilde{y} \mid \widetilde{a}; q_i) = \sum_{y_i} p(y_i, y_{-i} \mid \widetilde{a}) \prod_{\ell=0}^{L} q_i^{\ell}(\widetilde{y}_i^{\ell} \mid a_i, \widetilde{a}_i, y_i^0, \ldots, y_i^{\ell}, \widetilde{y}_i^0, \ldots, \widetilde{y}_i^{\ell-1}) \tag{2}$$

Given a distribution over action profiles α, the *detectability* of a manipulation (s_i, q_i) is the expected χ^2-divergence between the distribution of reports under obedience and truth-telling and that under the manipulation:

$$\chi_i^2(s_i, q_i, \alpha; p) = \sum_{a \in A} \sum_{\widetilde{a}_i \in A_i} \sum_{\widetilde{y} \in Y} \alpha(a) s_i(\widetilde{a}_i \mid a_i) p(\widetilde{y} \mid a) \left(\frac{p(\widetilde{y} \mid a) - p(\widetilde{y} \mid \widetilde{a}; q_i)}{p(\widetilde{y} \mid a)} \right)^2, \tag{3}$$

where $\widetilde{a} = (\widetilde{a}_i, a_{-i})$, the inner summand has the convention $c^2/0 = \infty$ for $c > 0$ and $0^2/0 = 0$, and I emphasize χ_i^2's dependence on the distribution p, which will vary when signals are lagged. Notice two aspects of detectability, as defined in (3). First, if q_i is truthful (i.e.,

[5]Because SW23 study the blind game in which the mediator observes the signals, only action manipulations are applicable to their setting. They consider the analogous one-shot deviation in which player i manipulates its actions in one period but is obedient in all other periods.

always places probability 1 on $\widetilde{y}_i = y_i$), then $p(\widetilde{y} \mid \widetilde{a}; q_i) = p(\widetilde{y} \mid \widetilde{a})$, and detectability is simply equal to the average χ^2-divergence between the signal distributions under a and under a', where the average is taken over $(a, a') \sim (\alpha, s_i)$. Second, if q_i places positive probability on a report profile $\widetilde{y} \notin \bar{Y}$—one the mediator knows to be impossible—then $p(\widetilde{y} \mid a) = 0$ while $p(\widetilde{y} \mid \widetilde{a}; q_i) > 0$, and $\chi_i^2(s_i, q_i, \alpha; p) = \infty$. This would arise, for example, if players publicly observe an aggregate signal; then different players having different observations of that signal has probability zero, and a message manipulation that does not always tell the truth is infinitely detectable.

The detectability of a deviation as defined in (3) will provide a bound that is too loose when part of the signal is observed on a delay. To see this intuitively, suppose that an entire signal is fully delayed by ℓ periods. Players are deterred from deviating by punishment. If that punishment comes ℓ periods later, then the size of the punishment decays by δ^ℓ in present-value terms, making deviation more profitable. This reasoning suggests that detectability should "discount" at the same rate as payoffs.

To formalize this intuition, let $p^{\le \ell}(\cdot \mid a)$ be the joint distribution of signals $y^0, \ldots, y^\ell$ conditional on an action profile a, and let $q_i^{\le \ell} = (q_i^{\ell'})_{\ell'=0}^{\ell}$. The *effective detectability* of a monitoring structure is the discount factor-weighted convex combination of cumulative detectabilities; that is,

$$\chi_{i,\text{eff}}^2(s_i, q_i, \alpha; p) = \delta^L \chi_i^2(s_i, q_i, \alpha; p) + (1-\delta) \sum_{\ell=0}^{L-1} \delta^\ell \chi_i^2(s_i, q_i^{\le \ell}, \alpha; p^{\le \ell}). \tag{4}$$

Let the *incremental detectability* in period ℓ be the difference between the cumulative detectabilities at lags ℓ and $\ell-1$: $\Delta\chi_i^2(s_i, q_i^{\le \ell}, \alpha; p^{\le \ell}) = \chi_i^2(s_i, q_i^{\le \ell}, \alpha; p^{\le \ell}) - \chi_i^2(s_i, q_i^{\le \ell-1}, \alpha; p^{\le \ell-1})$, with $\Delta\chi_i^2(s_i, q_i^0, \alpha; p^0) \equiv \chi_i^2(s_i, q_i^0, \alpha; p^0)$. Notice that $\Delta\chi_i^2(\cdot) \ge 0$: observing the lag-$\ell$ signal always increases detectability. We can also rewrite effective detectability as the discounted sum of incremental detectabilities,

$$\chi_{i,\text{eff}}^2(s_i, q_i, \alpha; p) = \sum_{\ell=0}^{L} \delta^\ell \Delta\chi_i^2(s_i, q_i^{\le \ell}, \alpha; p^{\le \ell}), \tag{5}$$

which will turn out to be useful for the analysis.

It will also be useful to consider, for a fixed action manipulation s_i, the message manipulation resulting in the smallest possible effective detectability. That is,

$$\underline{\chi}_{i,\text{eff}}^2(s_i, \alpha) = \min_{q_i} \chi_{i,\text{eff}}^2(s_i, q_i, \alpha; p). \tag{6}$$

This object is well-defined. To see this, note that the set of message manipulations is compact, the objective is bounded below by zero and continuous over values of q_i such that $\chi^2_{i,\text{eff}}(\cdot) < \infty$, and finite values are feasible because truth-telling always results in a finite $\chi^2_{i,\text{eff}}$ by the full-support assumption.

3 Strengthening the Sugaya–Wolitzky bound

This section contains the main result of the paper: a strengthening and extension to delayed signals of SW23's Theorem 1. Before doing so, I restate SW23's result. Restrict attention to an undelayed signal structure, and let $\chi_i^{2,T}(s_i, \alpha) = \chi_i^2(s_i, q_i^T, \alpha; p)$ be the detectability of an action manipulation with the truthful message manipulation. SW23's Theorem 1 shows that every Nash equilibrium occupation measure α of Γ for an undelayed signal satisfies

$$g_i(s_i, \alpha) \leq \sqrt{\frac{\delta}{1-\delta}\chi_i^{2,T}(s_i, \alpha)V_i(\alpha)} \quad \forall i, s_i. \tag{bound-SW23}$$

Theorem 1. *Every Nash equilibrium occupation measure α of Γ satisfies, for each player i, action manipulation s_i, and message manipulation q_i,*

$$g_i(s_i, \alpha) \leq \sqrt{\frac{\delta}{1-\delta}\,\chi^2_{i,\text{eff}}(s_i, q_i, \alpha; p)\,V_i(\alpha)}. \tag{7}$$

Minimizing the right-hand side over message manipulations yields

$$g_i(s_i, \alpha) \leq \sqrt{\frac{\delta}{1-\delta}\,\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha)\,V_i(\alpha)} \quad \forall i, s_i, \tag{bound}$$

and consequently α is a static ε-correlated equilibrium for

$$\varepsilon = \max_{i,s_i} \sqrt{\frac{\delta}{1-\delta}\,\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha)\,V_i(\alpha)}. \tag{8}$$

Several limit cases and comparative statics are immediate. When q_i is truthful, (7) is the effective-detectability bound for the true, undistorted signal. When in addition $\mathcal{L} = \{0\}$ (undelayed), it is exactly (bound-SW23). Because (bound) minimizes over messages, it is weakly sharper than both, while Proposition 1 demonstrates that it is bounded from below by the same expression with $\chi_i^2(\cdot; p_{-i})$. For a two-part signal with delay L, $\chi^2_{i,\text{eff}}$ approaches $\chi_i^2(\cdot; p^0)$—the detectability of the immediate signal—as $L \to \infty$. Lagging the signal affects incentives more as players become less patient and as the incremental detectability of the lagged signal over the contemporaneous signal increases.

Before going through the proof, I preview the intuition. The key step is Lemma 2, which has two parts. The first part bounds from below the expected variance of the period-$t+1$ continuation value, conditional on a history before period-t play, by the sum of the conditional variances of continuation payoffs explained by each signal observed between period t and period $t+1$ (which includes each signal with lag ℓ generated in period $t-\ell$). The intuition for this step is that because the revisions in expected continuations explained by each lagged signal are independent conditional on a period-t history, as the conditioning set includes the recommendations (which correspond to actions on path) that generated each signal, one can write the variance of continuation payoffs as the sum of the variances explained by each signal plus a nonnegative residual term capturing variation not explained by any individual signal.

The second part of Lemma 2 bounds from above the ratio of the squared gain from a deviation to the effective detectability of that deviation by the discounted sum of conditional variances of continuation payoffs explained by signals generated by period-t play. With an undelayed signal, deviating in period t and then adhering to the mediator's recommendations in all future periods causes a change to continuation payoffs equal to the sum over future histories of the difference in probabilities of reaching that history times the continuation payoff of that history, which must be no less than the period-t gain from deviation in equilibrium. SW23 then use the Cauchy–Schwarz inequality to bound this sum of products by the square root of the detectability times the variance of period-$t+1$ continuation payoffs. Here, with a lagged signal, deviation implies changes in continuation payoffs in periods $t+1, \ldots, t+L+1$ as more of the signal gets revealed. Incentive compatibility requires that the period-t gain from a deviation must be no more than the discounted sum of expected changes to continuation payoffs from the lagged signal observed ℓ periods later. Applying the Cauchy–Schwarz inequality separately for each ℓ bounds this quantity by the incremental detectability multiplied by the variance of continuation payoffs explained by the signal with that lag, and applying Cauchy–Schwarz again combines this expression into the discounted sum of incremental detectabilities—which is the root of the effective detectability—times the discounted sum of explained continuation variances. Because these explained continuation variances are the same objects from the first part of Lemma 2, just in a different order, applying these bounds for infinitely many periods yields a relation between effective detectability, gain, and payoff variance.

It is also worth pointing out that the shift from the blind game to the communication game—which underlies the tightening relative to SW23—does not add any complexity to the proof; all the new technical steps come from the extension to lagged monitoring structures. The key difference between the blind game and the communication game is that the mediator receives verified signals in the former but not the latter. However, because in equilibrium the mediator receives truthful reports of signals, the mediator acts as though the reports it receives are the true signals throughout. Letting players misreport signals strictly increases the set of potential deviations that are considered, tightening the bound, while the statement of Theorem 1 being a necessary condition means that the response to off-equilibrium path messages does not need to be specified, beyond the assertion that they make deviation unprofitable.

3.1 Proof of Theorem 1

Fix a Nash equilibrium outcome μ of Γ, which is a truthful and obedient equilibrium outcome of Γ^C. Throughout the proof, restrict attention to message manipulations whose reports lie in $\bar{Y}$ almost surely; any other manipulation has $\chi^2_{i,\text{eff}} = \infty$ by convention, so (7) holds trivially. Along a path of action profiles $a_1, a_2, \dots$ with positive probability under μ, let $u_i^t = u_i(a_t)$, and let w_i^t be player i's continuation payoff; that is,

$$w_i^t = (1-\delta) \sum_{t'=t}^{\infty} \delta^{t'-t} u_i^{t'}.$$

Let a history of actions and reported signals (which correspond to true signals on-path) at the beginning of period t be denoted by $h^t = \left(a_{t'}, (y^\ell_{t'-\ell})_{\ell=0}^{\min\{t'-1,L\}}\right)_{t'=1}^{t-1}$. Let $U_i^t = \mathbb{E}[u_i^t \mid h^t]$ and $W_i^t = \mathbb{E}[w_i^t \mid h^t]$ be the expected period-t utility and continuation payoff given history h^t. Noting that $w_i^t = (1-\delta)u_i^t + \delta w_i^{t+1}$, taking expectations with respect to h^t yields

$$W_i^t = (1-\delta)\, U_i^t + \delta \mathbb{E}[W_i^{t+1} \mid h^t]. \tag{9}$$

The first goal is to rewrite incremental detectabilities in terms of conditional and marginal likelihood ratios. Fix a player i, an action profile a, a manipulated action for player i $\widetilde{a}_i$, and a message manipulation q_i. Define the joint and incremental likelihood ratios of cumulative signal reports through lag ℓ as

$$\nu^{\le\ell}(\widetilde{y}^{\le\ell}; a, \widetilde{a}_i, q_i^{\le\ell}) = \frac{p(\widetilde{y}^{\le\ell} \mid \widetilde{a}_i, a_{-i}; q_i^{\le\ell})}{p(\widetilde{y}^{\le\ell} \mid a)}$$
$$\nu^{\le\ell|\ell-1}(\widetilde{y}^{\le\ell}; a, \widetilde{a}_i, q_i^{\le\ell}) = \frac{p(\widetilde{y}^{\le\ell} \mid \widetilde{y}^{\le\ell-1}, \widetilde{a}_i, a_{-i}; q_i^{\le\ell})}{p(\widetilde{y}^{\le\ell} \mid \widetilde{y}^{\le\ell-1}, a)}$$

The numerator of $\nu^{\le\ell}$ is the joint probability of the mediator observing signal reports $\widetilde{y}^{\le\ell}$ assuming that actions $\widetilde{a}$ were played, that player i is manipulating its messages according to q_i, and that other players are truthful, while the denominator is the probability under a

assuming all players are truthful. Letting $\overline{\nu}^{\times}(\cdot) = 1 - \nu^{\times}(\cdot)$, the cumulative detectability of a signal can be written as

$$\chi_i^2(s_i, q_i^{\leq \ell}, \alpha; p^{\leq \ell}) = \sum_{a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}} \alpha(a) s_i(\widetilde{a}_i \mid a_i) p(\widetilde{y}^{\leq \ell} \mid a) \overline{\nu}^{\leq \ell}(\widetilde{y}^{\leq \ell}; a, \widetilde{a}_i, q_i^{\leq \ell})^2. \tag{10}$$

Note also that (a) by construction $\nu^{\leq \ell} = \nu^{\leq \ell - 1} \nu^{\leq \ell | \ell - 1}$, and (b) since $p(\cdot \mid \widetilde{y}^{\leq \ell - 1}, \widetilde{a}; q_i^{\leq \ell})$ is a probability distribution, $\mathbb{E}_{p(\cdot | \widetilde{y}^{\leq \ell - 1}, a)}[\nu^{\leq \ell | \ell - 1}(\widetilde{y}^{\leq \ell}; a, \widetilde{a}_i, q_i^{\leq \ell})] = 1$.

We start by rewriting the incremental detectability in a form that will ultimately prove to be convenient.

Lemma 1. *For each ℓ, the incremental detectability of the lag-ℓ report satisfies*

$$\Delta \chi_i^2(s_i, q_i^{\leq \ell}, \alpha; p^{\leq \ell}) = \sum_{a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}} \alpha(a) s_i(\widetilde{a}_i \mid a_i) p(\widetilde{y}^{\leq \ell} \mid a) \times \left(\nu^{\leq \ell - 1}(\widetilde{y}^{\leq \ell - 1}; a, \widetilde{a}_i, q_i^{\leq \ell - 1}) \overline{\nu}^{\leq \ell | \ell - 1}(\widetilde{y}^{\leq \ell}; a, \widetilde{a}_i, q_i^{\leq \ell})\right)^2.$$

Proof. Fix $a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}, q_i$ and suppress them as arguments to $\nu^{\times}$ and $\overline{\nu}^{\times}$. Since $\nu^{\leq \ell} = \nu^{\leq \ell - 1} \nu^{\leq \ell | \ell - 1}$, we have

$$\nu^{\leq \ell - 1} \overline{\nu}^{\leq \ell | \ell - 1} = \nu^{\leq \ell - 1}(1 - \nu^{\leq \ell | \ell - 1}) = \nu^{\leq \ell - 1} - \nu^{\leq \ell} = \overline{\nu}^{\leq \ell} - \overline{\nu}^{\leq \ell - 1}.$$

Squaring and taking expectations over $\widetilde{y}^{\leq \ell} \sim p(\cdot \mid a)$,

$$\mathbb{E}_{p(\cdot|a)}\left[\left(\nu^{\leq \ell - 1} \overline{\nu}^{\leq \ell | \ell - 1}\right)^2\right] = \mathbb{E}_{p(\cdot|a)}\left[\left(\overline{\nu}^{\leq \ell}\right)^2 - 2 \overline{\nu}^{\leq \ell} \overline{\nu}^{\leq \ell - 1} + \left(\overline{\nu}^{\leq \ell - 1}\right)^2\right].$$

By the law of iterated expectations,

$$\mathbb{E}_{p(\cdot|a)}\left[\overline{\nu}^{\leq \ell} \overline{\nu}^{\leq \ell - 1}\right] = \mathbb{E}_{p(\cdot|a)}\left[\overline{\nu}^{\leq \ell - 1} \mathbb{E}_{p(\cdot | \widetilde{y}^{\leq \ell - 1}, a)}\left[\overline{\nu}^{\leq \ell} \middle| \widetilde{y}^{\leq \ell - 1}\right]\right] = \mathbb{E}_{p(\cdot|a)}\left[\left(\overline{\nu}^{\leq \ell - 1}\right)^2\right]$$

where the last equality uses $\mathbb{E}[\overline{\nu}^{\leq \ell} \mid \widetilde{y}^{\leq \ell - 1}] = 1 - \nu^{\leq \ell - 1} \mathbb{E}[\nu^{\leq \ell | \ell - 1} \mid \widetilde{y}^{\leq \ell - 1}] = 1 - \nu^{\leq \ell - 1} = \overline{\nu}^{\leq \ell - 1}$. Substituting, $\mathbb{E}_{p(\cdot|a)}\left[\left(\nu^{\leq \ell - 1} \overline{\nu}^{\leq \ell | \ell - 1}\right)^2\right] = \mathbb{E}_{p(\cdot|a)}\left[\left(\overline{\nu}^{\leq \ell}\right)^2\right] - \mathbb{E}_{p(\cdot|a)}\left[\left(\overline{\nu}^{\leq \ell - 1}\right)^2\right]$. Taking expectations over $a \sim \alpha$ and $\widetilde{a}_i \sim s_i(\cdot \mid a)$ turns these terms into $\chi_i^2(s_i, q_i^{\leq \ell}, \alpha; p^{\leq \ell})$ and $\chi_i^2(s_i, q_i^{\leq \ell - 1}, \alpha; p^{\leq \ell - 1})$; their difference is $\Delta \chi_i^2(s_i, q_i^{\leq \ell}, \alpha; p^{\leq \ell})$ by definition. □

We now prove the analogue of SW23's Lemma 1 for the delayed signal. Let β_t^{ℓ} be the expected variance in future payoffs in period $t + \ell + 1$ explained by the mediator receiving the message $\widetilde{y}_t^\ell$, which is generated by actions in period t and observed in period $t+\ell$. That is,

$$\beta_t^\ell = \mathbb{E}\left[\mathrm{Var}\left(\mathbb{E}\left[W_i^{t+\ell+1} \middle| h^{t+\ell}, \widetilde{y}_t^\ell\right] \middle| h^{t+\ell}\right)\right],$$

with convention $\beta_t^\ell = 0$ for $t \leq 0$. We show that (a) the variance in continuation payoffs explained by signals observed after period t is bounded above by the variance of continuation payoffs given the previous history, and (b) the variance in continuation payoffs explained by the signals generated by period-t play is bounded from below with the ratio of the squared gain from the deviation to the effective detectability of the deviation.

Lemma 2. *For all periods t, players i, action manipulations s_i, and message manipulations q_i, we have*

$$\sum_{\ell=0}^{L} \beta_{t-\ell}^\ell \leq \mathbb{E}[\mathrm{Var}(W_i^{t+1} \mid h^t)], \tag{11}$$

$$\sum_{\ell=0}^{L} \delta^\ell \beta_t^\ell \geq \left(\tfrac{1-\delta}{\delta}\right)^2 \frac{g_i(s_i, \alpha_t^\mu)^2}{\chi_{i,\text{eff}}^2(s_i, q_i, \alpha_t^\mu; p)}. \tag{12}$$

Proof. Inequality (11). The messages the mediator receives at the transition into period $t+1$ are $(\widetilde{y}_{t-\ell}^\ell)_{\ell=0}^{\min\{t-1,L\}}$. Let $D_{t-\ell}^\ell = \mathbb{E}[W_i^{t+1} \mid h^t, \widetilde{y}_{t-\ell}^\ell] - \mathbb{E}[W_i^{t+1} \mid h^t]$ be the change to player i's expected continuation value upon the mediator receiving $\widetilde{y}_{t-\ell}^\ell$. Note that (a) $\mathbb{E}[(D_{t-\ell}^\ell)^2] = \beta_{t-\ell}^\ell$; (b) $\mathbb{E}[D_{t-\ell}^\ell \mid h^t] = 0$; (c) because signals generated by actions in different periods are drawn independently conditional on actions in those periods, the on-path actions are determined by h^t, and on-path messages are truthful reports of those signals, $\mathbb{E}[D_{t-\ell}^\ell D_{t-\ell'}^{\ell'} \mid h^t] = \mathbb{E}[D_{t-\ell}^\ell \mid h^t]\mathbb{E}[D_{t-\ell'}^{\ell'} \mid h^t] = 0$ for $\ell \neq \ell'$; and (d) because $D_{t-\ell}^\ell$ is a conditional expectation of $W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t]$ given coarser information, $\mathbb{E}[D_{t-\ell}^\ell(W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t])] = \mathbb{E}[(D_{t-\ell}^\ell)^2] = \beta_{t-\ell}^\ell$ for all ℓ. Letting $t(L) \equiv \min\{t-1, L\}$, facts (d) and (c) imply that

$$\mathbb{E}\left[(W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t])\left(\sum_{\ell=0}^{t(L)} D_{t-\ell}^\ell\right)\right] = \mathbb{E}\left[\sum_{\ell=0}^{t(L)} \left(D_{t-\ell}^\ell\right)^2\right] = \mathbb{E}\left[\left(\sum_{\ell=0}^{t(L)} D_{t-\ell}^\ell\right)^2\right],$$

so the residual $W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t] - \sum_\ell D_{t-\ell}^\ell$ is orthogonal to $\sum_\ell D_{t-\ell}^\ell$. Therefore,

$$\mathbb{E}[\mathrm{Var}(W_i^{t+1} \mid h^t)] = \mathbb{E}\left[\left(\sum_{\ell=0}^{t(L)} D_{t-\ell}^\ell\right)^2\right] + \mathbb{E}\left[\left(W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t] - \sum_{\ell=0}^{t(L)} D_{t-\ell}^\ell\right)^2\right]$$

$$
\begin{aligned}
&= \sum_{\ell=0}^{t(L)} \beta_{t-\ell}^{\ell} + \mathbb{E}\left[\left(W_i^{t+1} - \mathbb{E}[W_i^{t+1} \mid h^t] - \sum_{\ell=0}^{t(L)} D_{t-\ell}^{\ell}\right)^2\right] \\
&\geq \sum_{\ell=0}^{t(L)} \beta_{t-\ell}^{\ell}.
\end{aligned}
$$

which, as $\beta_{t-\ell}^{\ell} = 0$ for $t \leq \ell$ by convention, gives (11).

Inequality (12). For $k \geq 1$, let $\overline{W}_{t+k}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) = \mathbb{E}[w_i^{t+k} \mid h^t, a_t = a, \widetilde{y}_t^{\leq \ell} = \widetilde{y}^{\leq \ell}]$ be player i's expected continuation payoff from period $t+k$ onward given the mediator's eventual receipt of the period-t reports through lag ℓ, $\widetilde{y}^{\leq \ell} = (\widetilde{y}^{\ell'})_{\ell'=0}^{\ell}$, and let $\overline{W}_{t+1}(\cdot) = \overline{W}_{t+1}^{\leq L}(\cdot)$ be the ultimate expected continuation payoff after all signals are realized. Note that $\overline{W}_{t+1}$ is the deviator's expected continuation payoff under the deviation conditional on the reports: given the recommendation a and reports $\widetilde{y}$, the distribution of future recommendations—and hence, by obedience and truthfulness from period $t+1$ on, of the action path that determines w_i^{t+1}—is the same as on path, so the deviator's privately observed action and signals affect play only through the reports.

Consider the following one-shot deviation for player i: apply action manipulation s_i in period t and message manipulation q_i to all signals generated by period-t play, and remain obedient and truthful in all other cases. As μ is an equilibrium outcome, this deviation is unprofitable. Unprofitability of the deviation implies that

$$
\frac{1-\delta}{\delta} g_i(\alpha_t^{\mu}) \leq \sum_{h^t, a, \widetilde{a}_i} \mu(h^t, a) s_i(\widetilde{a}_i \mid a_i) \underbrace{\sum_{\widetilde{y}} \left(p(\widetilde{y} \mid a) - p(\widetilde{y} \mid \widetilde{a}; q_i)\right) \overline{W}_{t+1}(h^t, a, \widetilde{y})}_{\equiv \Delta W_{t+1}(h^t, a, \widetilde{a}_i, q_i)}, \tag{13}
$$

where $\widetilde{a} = (\widetilde{a}_i, a_{-i})$, $\mu(\cdot)$ is the on-path probability of (h^t, a), and the dependence of $g_i(\cdot)$ on s_i is suppressed. $\Delta W_{t+1}(h^t, a, \widetilde{a}_i, q_i)$ is the change to expected continuation payoffs, taking expectations over the signal realizations conditional on the actions, from playing action manipulation s_i and message manipulation q_i. Notice that $p(\widetilde{y} \mid a) - p(\widetilde{y} \mid \widetilde{a}; q_i) = p(\widetilde{y} \mid a)\overline{\nu}^{\leq L}(\widetilde{y}; a, \widetilde{a}_i, q_i)$, and

$$
\overline{\nu}^{\leq L} = \sum_{\ell=0}^{L} \left(\overline{\nu}^{\leq \ell} - \overline{\nu}^{\leq \ell - 1}\right) = \sum_{\ell=0}^{L} \nu^{\leq \ell-1} \overline{\nu}^{\leq \ell \mid \ell - 1}.
$$

where we define $\nu^{\leq -1} = 1$ and $\overline{\nu}^{\leq -1} = 0$. Further, notice that

$$
\overline{W}_{t+1}^{\leq \ell-1}(h^t, a, \widetilde{y}^{\leq \ell-1}) = \mathbb{E}\left[w_i^{t+1} \mid h^t, a, \widetilde{y}^{\leq \ell-1}\right]
$$

$$= \mathbb{E}\left[\mathbb{E}\left[w_i^{t+1} \mid h^t, a, \widetilde{y}^{\leq \ell}\right] \mid h^t, a, \widetilde{y}^{\leq \ell-1}\right]$$
$$= \mathbb{E}\left[\overline{W}_{t+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) \mid h^t, a, \widetilde{y}^{\leq \ell-1}\right].$$

Finally, $\sum_{\widetilde{y}^{\leq \ell}} p(\widetilde{y}^{\leq \ell} \mid a)\nu^{\leq \ell-1}\overline{\nu}^{\leq \ell|\ell-1}\overline{W}_{t+1}^{\leq \ell-1} = 0$: $\nu^{\leq \ell-1}$ and $\overline{W}_{t+1}^{\leq \ell-1}$ factor out of the sum over $\widetilde{y}^{\ell}$, and the two remaining conditional distributions each sum to one. Thus, we can rewrite

$$\Delta W_{t+1}(h^t, a, \widetilde{a}_i, q_i) = \sum_{\ell=0}^{L} \Sigma^{\leq \ell}(h^t, a, \widetilde{a}_i, q_i), \tag{14}$$
$$\Sigma^{\leq \ell}(h^t, a, \widetilde{a}_i, q_i) = \sum_{\widetilde{y}^{\leq \ell}} p(\widetilde{y}^{\leq \ell} \mid a)\nu^{\leq \ell-1}\overline{\nu}^{\leq \ell|\ell-1}\left(\overline{W}_{t+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) - \overline{W}_{t+1}^{\leq \ell-1}(h^t, a, \widetilde{y}^{\leq \ell-1})\right),$$

where the dependence of $\nu^{\leq \ell-1}$ and $\overline{\nu}^{\leq \ell|\ell-1}$ on $(\widetilde{y}^{\leq \ell}, a, \widetilde{a}_i, q_i)$ is suppressed, and $\overline{W}_{t+1}^{\leq -1}(h^t, a) = \mathbb{E}_{y^0 \sim p(\cdot|a)}[\overline{W}_{t+1}^{\leq 0}(h^t, a, y^0)]$. Finally, because $\widetilde{y}_t^{\ell}$ is received after period $t+\ell$, and hence only affects continuation payoffs starting in period $t+\ell+1$, we can write $\overline{W}_{t+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) = (1-\delta)\sum_{t'=t+1}^{t+\ell} \delta^{t'-t-1}\mathbb{E}[u_i^{t'} \mid h^t, a, \widetilde{y}^{\leq \ell}] + \delta^{\ell}\,\overline{W}_{t+\ell+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell})$, where $\mathbb{E}[u_i^{t'} \mid h^t, a, \widetilde{y}^{\leq \ell}] = \mathbb{E}[u_i^{t'} \mid h^t, a, \widetilde{y}^{\leq \ell-1}]$ for $t' \leq t+\ell$ by the independence argument below. Differencing, we have

$$\Sigma^{\leq \ell}(\cdot) = \delta^{\ell} \sum_{\widetilde{y}^{\leq \ell}} p(\widetilde{y}^{\leq \ell} \mid a)\nu^{\leq \ell-1}\overline{\nu}^{\leq \ell|\ell-1}\left(\overline{W}_{t+\ell+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) - \overline{W}_{t+\ell+1}^{\leq \ell-1}(h^t, a, \widetilde{y}^{\leq \ell-1})\right).$$

One can interpret $\Sigma^{\leq \ell}$ as the component of the expected change in continuation payoffs from the deviation that operates through the mediator's receipt of the lag-ℓ report. This report can only affect continuation payoffs starting in period $t+\ell+1$, at which point the change to period-$(t+1)$ continuation payoffs is discounted by δ^{ℓ}.

Letting $m(h^t, a, \widetilde{a}_i) = \mu(h^t, a)s_i(\widetilde{a}_i \mid a_i)$ be the probability of history h^t, action a being recommended, and manipulated action $\widetilde{a}_i$ being played in period t, we therefore have

$$\begin{aligned}
&\sum_{h^t, a, \widetilde{a}_i} m(h^t, a, \widetilde{a}_i)\Sigma^{\leq \ell}(h^t, a, \widetilde{a}_i, q_i) \\
&\quad = \delta^{\ell} \sum_{h^t, a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}} m(h^t, a, \widetilde{a}_i)p(\widetilde{y}^{\leq \ell} \mid a)\nu^{\leq \ell-1}\overline{\nu}^{\leq \ell|\ell-1}\left(\overline{W}_{t+\ell+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) - \overline{W}_{t+\ell+1}^{\leq \ell-1}(h^t, a, \widetilde{y}^{\leq \ell-1})\right) \\
&\quad \leq \delta^{\ell} \sqrt{\sum_{h^t, a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}} m(h^t, a, \widetilde{a}_i)p(\widetilde{y}^{\leq \ell} \mid a)\left(\nu^{\leq \ell-1}\overline{\nu}^{\leq \ell|\ell-1}\right)^2} \times \\
&\qquad \sqrt{\sum_{h^t, a, \widetilde{a}_i, \widetilde{y}^{\leq \ell}} m(h^t, a, \widetilde{a}_i)p(\widetilde{y}^{\leq \ell} \mid a)\left(\overline{W}_{t+\ell+1}^{\leq \ell}(h^t, a, \widetilde{y}^{\leq \ell}) - \overline{W}_{t+\ell+1}^{\leq \ell-1}(h^t, a, \widetilde{y}^{\leq \ell-1})\right)^2} \\
&\quad = \delta^{\ell} \sqrt{\Delta\chi_i^2(s_i, q_i^{\leq \ell}, \alpha_t^{\mu}; p^{\leq \ell})\mathbb{E}\left[\mathrm{Var}\left(\overline{W}_{t+\ell+1}^{\leq \ell} \mid h^t, a, \widetilde{y}^{\leq \ell-1}\right)\right]}
\end{aligned}$$

$$\leq \delta^\ell \sqrt{\Delta\chi_i^2(s_i, q_i^{\leq \ell}, \alpha_t^\mu; p^{\leq \ell}) \mathbb{E}\left[\operatorname{Var}\left(\mathbb{E}\left[W_i^{t+\ell+1} \middle| h^{t+\ell}, \widetilde{y}_t^\ell\right] \middle| h^{t+\ell}\right)\right]}$$
$$= \delta^\ell \sqrt{\Delta\chi_i^2(s_i, q_i^{\leq \ell}, \alpha_t^\mu; p^{\leq \ell}) \beta_t^\ell}.$$

The first inequality is Cauchy–Schwarz. The second equality applies Lemma 1 to the first term and, to the second, the fact that $\overline{W}_{t+\ell+1}^{\leq \ell-1}$ is the conditional mean of $\overline{W}_{t+\ell+1}^{\leq \ell}$, as above. The second inequality holds by the following argument. Because signals are drawn independently across periods conditional on actions, and $h^{t+\ell}$ includes $(a_t, \widetilde{y}_t^{\leq \ell-1})$ but not $\widetilde{y}_t^\ell$, the report $\widetilde{y}_t^\ell$ is independent of $h^{t+\ell}$ given $(h^t, a, \widetilde{y}_t^{\leq \ell-1})$. Thus $\overline{W}_{t+\ell+1}^{\leq \ell}$ averages the finer conditional expectation $\mathbb{E}[W_i^{t+\ell+1} \mid h^{t+\ell}, \widetilde{y}_t^\ell]$ over intervening histories $h^{t+\ell}$ with weights that do not depend on $\widetilde{y}_t^\ell$. As the variance of an average is at most the average of the variances, averaging over the intervening history cannot raise the variance of the continuation payoff explained by $\widetilde{y}_t^\ell$.

Combining this inequality with (13) and (14) gives

$$\begin{aligned}
\frac{1-\delta}{\delta} g_i(\alpha_t^\mu) &\leq \sum_{h^t, a, \widetilde{a}_i} m(h^t, a, \widetilde{a}_i) \Delta W_{t+1}(h^t, a, \widetilde{a}_i, q_i) \\
&= \sum_{\ell=0}^{L} \sum_{h^t, a, \widetilde{a}_i} m(h^t, a, \widetilde{a}_i) \Sigma^{\leq \ell}(h^t, a, \widetilde{a}_i, q_i) \\
&\leq \sum_{\ell=0}^{L} \delta^\ell \sqrt{\Delta\chi_i^2(s_i, q_i^{\leq \ell}, \alpha_t^\mu; p^{\leq \ell}) \beta_t^\ell} \\
&\leq \sqrt{\left(\sum_{\ell=0}^{L} \delta^\ell \Delta\chi_i^2(s_i, q_i^{\leq \ell}, \alpha_t^\mu; p^{\leq \ell})\right)\left(\sum_{\ell=0}^{L} \delta^\ell \beta_t^\ell\right)} \\
&= \sqrt{\chi_{i,\mathrm{eff}}^2(s_i, q_i, \alpha_t^\mu; p)\left(\sum_{\ell=0}^{L} \delta^\ell \beta_t^\ell\right)}
\end{aligned}$$

where the second inequality is another application of Cauchy–Schwarz. Squaring and rearranging gives (12).[6] □

[6]Strictly speaking, squaring is valid only when $g_i(s_i, \alpha_t^\mu) \geq 0$; the deviation may have a negative flow gain in some periods, in which case (12) should be read with $\max\{g_i(s_i, \alpha_t^\mu), 0\}^2$ in the numerator (the same implicit step appears in SW23's Lemma 1). This does not affect the result, as $(x, z) \mapsto \max\{x, 0\}^2/z$ is convex for $z > 0$, so the Jensen step below goes through with $\max\{g_i(\cdot), 0\}^2$ in place of $g_i(\cdot)^2$ throughout, and (7) follows because $g_i(s_i, \alpha^\mu) \leq \max\{g_i(s_i, \alpha^\mu), 0\}$. This form also guarantees that each α_t^μ lies in the domain of convexity used below: by the unsquared inequality above, $\chi_{i,\mathrm{eff}}^2(s_i, q_i, \alpha_t^\mu; p) = 0$ implies $g_i(s_i, \alpha_t^\mu) \leq 0$, and hence $\max\{g_i(s_i, \alpha_t^\mu), 0\} = 0$.

Now we recursively bound the payoff variance. By SW23's Lemma 1,

$$\mathrm{Var}(\mathbb{E}[W_i^{T+1} \mid h^T]) \geq \frac{1}{\delta}\mathrm{Var}(W_i^T) - \frac{1-\delta}{\delta}\mathrm{Var}(U_i^T). \tag{15}$$

By the law of total variance, $\mathrm{Var}(W_i^{T+1}) = \mathrm{Var}(\mathbb{E}[W_i^{T+1} \mid h^T]) + \mathbb{E}[\mathrm{Var}(W_i^{T+1} \mid h^T)]$. Therefore, combining (15) and (11), we have

$$\mathrm{Var}(W_i^{T+1}) \geq \frac{1}{\delta}\mathrm{Var}(W_i^T) - \frac{1-\delta}{\delta}\mathrm{Var}(U_i^T) + \sum_{\ell=0}^{L} \beta_{T-\ell}^{\ell}.$$

Recursively applying this inequality over $t = 1, \ldots, T$, multiplying both sides by δ^T, and canceling $\mathrm{Var}(W_i^1) = 0$ yields

$$\delta^T \mathrm{Var}(W_i^{T+1}) \geq \sum_{t=1}^{T} \delta^t \sum_{\ell=0}^{L} \beta_{t-\ell}^{\ell} - (1-\delta) \sum_{t=1}^{T} \delta^{t-1} \mathrm{Var}(U_i^t).$$

Take limits as $T \to \infty$. Payoffs are bounded, hence $\mathrm{Var}(W_i^{T+1})$ is bounded; therefore the left-hand side of the above expression vanishes. Rearranging gives

$$\begin{aligned}
(1-\delta) \sum_{t=1}^{\infty} \delta^{t-1} \mathrm{Var}(U_i^t) &\geq \sum_{t=1}^{\infty} \delta^t \sum_{\ell=0}^{L} \beta_{t-\ell}^{\ell} \\
&= \sum_{t=1}^{\infty} \delta^t \sum_{\ell=0}^{L} \delta^{\ell} \beta_t^{\ell} \\
&\geq \sum_{t=1}^{\infty} \delta^t \left(\frac{1-\delta}{\delta}\right)^2 \frac{g_i(s_i, \alpha_t^{\mu})^2}{\chi_{i,\mathrm{eff}}^2(s_i, q_i, \alpha_t^{\mu}; p)}
\end{aligned}$$

where the last inequality is (12). We therefore have

$$\delta \sum_{t=1}^{\infty} \delta^{t-1} \mathrm{Var}(U_i^t) \geq (1-\delta) \sum_{t=1}^{\infty} \delta^{t-1} \frac{g_i(s_i, \alpha_t^{\mu})^2}{\chi_{i,\mathrm{eff}}^2(s_i, q_i, \alpha_t^{\mu}; p)}. \tag{16}$$

The function $f_i(\alpha) \equiv \frac{g_i(s_i,\alpha)^2}{\chi_{i,\mathrm{eff}}^2(s_i,q_i,\alpha;p)}$ is convex in α over the set $\{\alpha \in \Delta(A) : \chi_{i,\mathrm{eff}}^2(s_i, q_i, \alpha; p) = 0 \implies g_i(s_i, \alpha) = 0\}$, with convention $0/0 = 0$. The proof is identical to SW23's Lemma 2, as $g_i(\cdot)$ and $\chi_{i,\mathrm{eff}}^2(\cdot)$ are both linear in α, given a fixed q_i. Therefore, Jensen's inequality implies that

$$(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\frac{g_i(s_i,\alpha_t^\mu)^2}{\chi^2_{i,\text{eff}}(s_i,q_i,\alpha_t^\mu;p)} \geq \frac{g_i\left(s_i,(1-\delta)\sum_t \delta^{t-1}\alpha_t^\mu\right)^2}{\chi^2_{i,\text{eff}}\left(s_i,q_i,(1-\delta)\sum_t \delta^{t-1}\alpha_t^\mu;p\right)} = \frac{g_i(s_i,\alpha^\mu)^2}{\chi^2_{i,\text{eff}}(s_i,q_i,\alpha^\mu;p)}. \tag{17}$$

In addition, as proved by SW23,

$$\frac{\delta}{1-\delta}V_i(\alpha^\mu) \geq \delta\sum_{t=1}^{\infty}\delta^{t-1}\text{Var}(U_i^t). \tag{18}$$

Substituting (17) and (18) into (16) gives

$$\frac{\delta}{1-\delta}V_i(\alpha^\mu) \geq \frac{g_i(s_i,\alpha^\mu)^2}{\chi^2_{i,\text{eff}}(s_i,q_i,\alpha^\mu;p)}.$$

Rearranging and taking square roots gives (7). Minimizing the right-hand side of (7) over q_i gives (bound), while maximizing $g_i(\cdot)/\sqrt{\underline{\chi}^2_{i,\text{eff}}(\cdot)}$ over s_i and i gives (8). □

4 Quantifying the improvement of this bound

(bound) tightens (bound-SW23) by allowing deviators to lie about their signals. This section analyzes when, and to what extent, this tightening is important. Intuitively, the tightening will be shown to be more important as rivals' signals become less informative of player i's true actions and signals, implying that this non-verifiability matters for games in which private information is first-order.

For exposition, I restrict attention to a no-delay signal; all results in this section have counterparts for delayed signal structures presented in Appendix A.1. I therefore drop the eff subscripts on $\underline{\chi}_i^2$ and $\chi_i^{2,T}$. Let q_i^T be the truthful message manipulation that reports the true signals with probability 1, and let $\chi_i^{2,T}(s_i,\alpha)=\chi_i^2(s_i,q_i^T,\alpha;p)$ be the detectability under a given manipulation with a truthful message. Since q_i^T is feasible in (6), it is immediate that $\underline{\chi}_i^2 \leq \chi_i^{2,T}$. The importance of signal non-verifiability is proportional to the strictness of this inequality. To characterize this strictness, we can use the following lemma to decompose χ_i^2 into the detectability of rivals' signals, which is unaffected by i's message manipulation, plus a nonnegative remainder term that depends on the manipulation.[7]

Lemma 3. *For any manipulation (s_i,q_i) and action distribution α, $\chi_i^2(s_i,q_i,\alpha;p)$ =*

[7]This is the order-two case of the chain rule for Rényi divergences (Polyanskiy and Wu, 2025, Section 7.12).

$\chi_i^2(s_i, \alpha; p_{-i}) + R(s_i, q_i, \alpha; p)$, *where*

$$\chi_i^2(s_i, \alpha; p_{-i}) = \sum_{a, \widetilde{a}_i, y_{-i}} \alpha(a) s_i(\widetilde{a}_i \mid a_i) p(y_{-i} \mid a) \left(\frac{p(y_{-i} \mid a) - p(y_{-i} \mid \widetilde{a})}{p(y_{-i} \mid a)} \right)^2$$

is independent of q_i, *and*

$$R(\cdot) = \sum_{a, \widetilde{a}_i, \widetilde{y}_i, y_{-i}} \alpha(a) s_i(\widetilde{a}_i \mid a_i) \frac{p(y_{-i} \mid \widetilde{a})^2}{p(y_{-i} \mid a)} p(\widetilde{y}_i \mid y_{-i}, a) \left(\frac{p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}; q_i) - p(\widetilde{y}_i \mid y_{-i}, a)}{p(\widetilde{y}_i \mid y_{-i}, a)} \right)^2 \geq 0$$

with $\widetilde{a} = (\widetilde{a}_i, a_{-i})$ *and* $p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}; q_i) = \sum_{y_i} q_i(\widetilde{y}_i \mid a_i, \widetilde{a}_i, y_i) p(y_i \mid y_{-i}, \widetilde{a})$ *the conditional distribution of i's report given rivals' signals under the manipulation.*

Proof. See Appendix A.2. □

Lemma 3 implies that $\chi_i^2(s_i, \alpha; p_{-i})$—the detectability of action manipulation s_i using only rivals' signals—is a lower bound for $\underline{\chi}_i^2(s_i, \alpha)$. Intuitively, this is because players $-i$ are truthful on path—the mediator always has access to the true y_{-i}—and player i's message can only provide additional information about $\widetilde{a}_i$. Thus, the best player i can hope for is to simply not add any additional information.

The following proposition, which is the main result of this section, shows that the minimum detectability varies in this range based on the extent to which player i can disguise its action with its report. Player i can fully disguise its action if the action that was recommended by the mediator is less informative in the Blackwell sense about its signal, conditional on its rivals' signals, than its deviation action. In this case, $\underline{\chi}_i^2$ attains its lower bound of the χ_i^2 of rivals $-i$. On the other hand, if varying i's action does not change the distribution of i's signals conditional on rivals' signals, then i's signal does not provide any additional information to the mediator, and truth-telling attains the lower bound.

Proposition 1. *Assume* $L = 0$, *and let* $\bar{Y}_{-i}$ *be the set of possible realizations of* y_{-i} *from* $\bar{Y}$. *For all* α *and* s_i,

$$\chi_i^2(s_i, \alpha; p_{-i}) \leq \underline{\chi}_i^2(s_i, \alpha) \leq \chi_i^{2,T}(s_i, \alpha). \tag{19}$$

In addition,

(i) $\chi_i^2(s_i, \alpha; p_{-i}) = \chi_i^{2,T}(s_i, \alpha)$*—and hence both are equal to* $\underline{\chi}_i^2(s_i, \alpha)$*—if and only if for all* $a \in \operatorname{supp} \alpha$, $\widetilde{a}_i \in \operatorname{supp} s_i(\cdot \mid a_i)$, *and* $y_{-i} \in \bar{Y}_{-i}$, $p(y_i \mid a, y_{-i}) = p(y_i \mid \widetilde{a}_i, a_{-i}, y_{-i})$;

(ii) $\underline{\chi}_i^2(s_i, \alpha) = \chi_i^2(s_i, \alpha; p_{-i})$ *if and only if, for each* a_i *in the support of* α*'s marginal and* $\widetilde{a}_i \in \operatorname{supp} s_i(\cdot \mid a_i)$, *the family* $\{p(\cdot \mid \widetilde{a}_i, a_{-i}, y_{-i})\}$ *dominates the family* $\{p(\cdot \mid a, y_{-i})\}$ *in the Blackwell order as experiments indexed by* $a_{-i} \in \operatorname{supp} \alpha(\cdot \mid a_i)$ *and* $y_{-i} \in \bar{Y}_{-i}$.

Proof. The first inequality in (19) is immediate from Lemma 3, while the second is immediate from the definition of $\underline{\chi}_i^2$ (6). To establish (i), note that $\chi_i^2(s_i, \alpha; p_{-i}) = \chi_i^{2,T}(s_i, \alpha)$ is equivalent to $R(s_i, q_i^T, \alpha; p) = 0$. Let $w(a, \widetilde{a}_i, y) = \alpha(a)s_i(\widetilde{a}_i \mid a_i)p(y_{-i} \mid \widetilde{a})^2 p(y_i \mid y_{-i}, a)/p(y_{-i} \mid a)$. The full-support assumption on p implies that, for $y \in \bar{Y}$, $w(a, \widetilde{a}_i, y) > 0$ if and only if $\alpha(a) > 0$ and $s_i(\widetilde{a}_i \mid a_i) > 0$. Then, we have

$$R(s_i, q_i^T, \alpha) = \sum_{a, \widetilde{a}_i, y} w(a, \widetilde{a}_i, y) \left(\frac{p(y_i \mid y_{-i}, \widetilde{a}_i, a_{-i}) - p(y_i \mid y_{-i}, a)}{p(y_i \mid y_{-i}, a)} \right)^2.$$

$R(s_i, q_i^T, \alpha; p) = 0$ if and only if the term inside the parentheses is zero for all $a, \widetilde{a}_i, y$ such that $w(a, \widetilde{a}_i, y) > 0$, which is true if and only if $p(y_i \mid y_{-i}, \widetilde{a}_i, a_{-i}) = p(y_i \mid y_{-i}, a)$ for all such $a, \widetilde{a}_i, y$. This establishes (i). For (ii), Blackwell dominance of the family of experiments is equivalent to each on-path conditional being a garbling of its deviation counterpart; that is, there exists a Markov kernel $q_i(\cdot \mid a_i, \widetilde{a}_i, y_i)$ such that for all $\widetilde{y}_i$, $a_{-i} \in \operatorname{supp} \alpha(\cdot \mid a_i)$, and $y_{-i} \in \bar{Y}_{-i}$,

$$p(\widetilde{y}_i \mid y_{-i}, a) = \sum_{y_i} p(y_i \mid y_{-i}, \widetilde{a}_i, a_{-i}) q_i(\widetilde{y}_i \mid a_i, \widetilde{a}_i, y_i) = p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}_i, a_{-i}; q_i).$$

If such a q_i exists, then using it as the message manipulation results in $R(s_i, q_i, \alpha; p) = 0$, attaining the lower bound. Otherwise, for all q_i, there exists an $a, \widetilde{a}_i, \widetilde{y}_i, y_{-i}$ such that $\alpha(a) > 0$, $s_i(\widetilde{a}_i \mid a_i) > 0$, $(\widetilde{y}_i, y_{-i}) \in \bar{Y}$, and $p(\widetilde{y}_i \mid y_{-i}, a) \neq p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}_i, a_{-i}; q_i)$, implying that $R(s_i, q_i, \alpha; p) > 0$ for all q_i, and therefore $\underline{\chi}_i^2(s_i, \alpha) > \chi_i^2(s_i, \alpha; p_{-i})$. □

To better parse the economic contexts in which $\underline{\chi}_i^2$ provides an improvement over $\chi_i^{2,T}$ or attains the lower bound, it is useful to provide sufficient conditions for each of the conditions in Proposition 1. Let a monitoring structure be *i-semipublic* if there is a function h such that for all $y \in \bar{Y}$, $y_i = h(y_{-i})$ almost surely. This is a weaker condition than public monitoring ($y_i = y_j$ for all i, j). i-semipublic monitoring is sufficient for message manipulation from player i to be ineffective, as i's signal is already observable from other signals available to the mediator. On the other hand, two conditions are sufficient for $\underline{\chi}_i^2$ to attain the lower bound. First, if i's signals are conditionally independent of rivals' signals given an action profile, and i can infer its rivals' action profile from the underlying signal distribution and i's recommendation, then i can attain the lower bound by drawing its signal report from the conditional distribution of its signal given the recommended action profile and observed signal. Second, if i's action enters its signal as a location shifter separate from rivals' signals, then i can fully "difference out" the deviation from the signal. These implications are collected in the following corollary.

Corollary 1. *Under i-semipublic monitoring,* $\chi_i^2(s_i, \alpha; p_{-i}) = \underline{\chi}_i^2(s_i, \alpha) = \chi_i^{2,T}(s_i, \alpha)$*. Meanwhile,* $\chi_i^2(s_i, \alpha; p_{-i}) = \underline{\chi}_i^2(s_i, \alpha)$ *if either of the following is true:*

(i) $y_i \perp y_{-i}$ *conditional on* a *and, for each* $a, a' \in \operatorname{supp} \alpha$*, if* $a_i = a_i'$*, then* $a = a'$*. In this case,* $q_i(\widetilde{y}_i \mid a_i, \widetilde{a}_i, y_i) = p(\widetilde{y}_i \mid a)$ *attains the lower bound, where* a *is the (unique) action profile featuring* a_i *with support in* α*.*

(ii) There exist functions g_i *and* g_{-i} *such that* $y_i = g_i(a_i) + g_{-i}(a_{-i}) + \zeta_i$*, where* ζ_i *is independent of* a *conditional on* y_{-i}*. In this case, reporting* $\widetilde{y}_i = y_i - g_i(\widetilde{a}_i) + g_i(a_i)$ *with probability 1 attains the lower bound.*

This result allows two remarks about the tightness of (bound). First, since in Γ rivals observe player i's action only through their own signals, one might expect an even tighter bound that uses $\chi_{i,\text{eff}}^2(s_i, \alpha; p_{-i})$, which is weakly smaller than $\underline{\chi}_{i,\text{eff}}^2$, to be valid. However, a bound based solely on $\chi_{i,\text{eff}}^2(\cdot; p_{-i})$ cannot be uniformly valid given the proof's strategy. Even if the marginal distribution of y_{-i} does not depend at all on a_i, and therefore rivals' signals say nothing about player i's true action, player i may still have an incentive to cooperate because she only observes her rivals' future behavior through her own signal. If player i gains from coordinating her actions with rival players' actions, then deviating can inhibit her ability to coordinate in the future, which disciplines that deviation. The case in which that is not true is exactly when player i's deviation action and signal are sufficiently informative of rivals' signals, which is the condition of Proposition 1(ii).[8]

Second, SW23's Theorems 2 and 3 demonstrate that their bound is tight in the low-discounting low-monitoring double limit for public monitoring games satisfying particular restrictions on the monitoring structure (pairwise identifiability or product structure monitoring). For such games, they prove that there exists a constant $k > 0$ such that cooperation is possible if $\min_{i,s_i} \chi_i^{2,T}(s_i, a)/(1-\delta) > k$ for all a, where $\chi_i^{2,T}(s_i, a)$ denotes detectability at the degenerate action distribution on a. Because this result provides a weakly tighter bound—$\underline{\chi}_i^2(\cdot) \leq \chi_i^{2,T}(\cdot)$—the same result applies for $\underline{\chi}_i^2(\cdot)$ as well. One may wonder whether using $\underline{\chi}_i^2$ implies that this result can be strengthened. However, because SW23 only prove this for public monitoring games and public monitoring implies i-semipublic monitoring for all i, Corollary 1 applies and $\underline{\chi}_i^2 = \chi_i^{2,T}$. This paper's bound therefore achieves exactly the same tightness as SW23's in these cases.

[8] AK19's bound uses the total variation distance of the rival-only signal distribution because they consider an information-free deviation in which player i plays an action a_i for every future period, and thereby foregoes any future payoffs from coordination. I discuss the relationship between my bound and AK19's further in Section 4.2.

4.1 Checking for a strict inequality

Proposition 1 does not provide a full characterization of when $\underline{\chi}_i^2 < \chi_i^{2,T}$. Truth-telling could be optimal, and hence $\underline{\chi}_i^2 = \chi_i^{2,T}$, for two reasons. The first is when there is nothing for i to hide—either because $a_i = \widetilde{a}_i$ with high probability according to s_i, or because a_{-i}, y_{-i} is highly informative of a_i, y_i, the distribution of $y_i \mid y_{-i}, \widetilde{a}_i, a_{-i}$ is exactly the distribution of $y_i \mid y_{-i}, a$, and $R(\cdot) = 0$ for the truthful manipulation. This case has a clean characterization captured in condition (i) of Proposition 1. The second is when $R(\cdot) > 0$ under truthful manipulation—there is some loss—but no additional gains from manipulating the signal distribution can be made because i cannot correlate the report to rivals' actions or signals. This occurs when neither the information sets of i nor the mediator dominate each other; a_{-i}, y_{-i} does not perfectly predict, nor is perfectly predicted by, a_i, y_i.

While clean assumptions on primitives that guarantee strict suboptimality of truth-telling are unavailable, it is still relatively straightforward to check whether truth-telling is optimal, and hence whether $\underline{\chi}_i^2(s_i, \alpha) = \chi_i^{2,T}(s_i, \alpha)$. Such a characterization is available because the minimization in (6) is a convex problem: by Lemma 3, the objective is a nonnegatively weighted sum of squared differences between the deviation and on-path conditional distributions of i's reported signal, the conditional distribution of the reported signal is linear in q_i, and the set of message manipulations is convex. Thus, first-order conditions are necessary and sufficient for optimality, and truth-telling is globally optimal if and only if the marginal benefit of shifting a small amount of weight away from the truth to another signal realization is nonpositive.

Fix a recommended action a_i and a played action $\widetilde{a}_i$ that arise with positive probability under (s_i, α), as well as a message manipulation q_i. Define the *excess score* of a report $\widetilde{y}_i$ given a received signal y_i as

$$\Lambda(\widetilde{y}_i \mid y_i) = \sum_{a_{-i}, y_{-i}} \alpha(a_{-i} \mid a_i) \frac{p(y_{-i} \mid \widetilde{a})^2}{p(y_{-i} \mid a)} p(y_i \mid y_{-i}, \widetilde{a}) \left(\frac{p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}; q_i) - p(\widetilde{y}_i \mid y_{-i}, a)}{p(\widetilde{y}_i \mid y_{-i}, a)} \right), \quad (20)$$

where $\alpha(a_{-i} \mid a_i)$ is the conditional distribution of rivals' recommendations given i's, $\widetilde{a} = (\widetilde{a}_i, a_{-i})$, $p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}; q_i)$ is the conditional distribution of i's report given rivals' signals from Lemma 3, and the dependence on $(a_i, \widetilde{a}_i, q_i)$ is suppressed.[9] The final term in large parentheses is the excess probability with which the manipulation produces report $\widetilde{y}_i$ compared to on-path play, conditional on rivals' signals. This is summed over (a_{-i}, y_{-i}) with weights equal to their conditional probability given (a_i, y_i), tilted by the same likelihood ratios that appear in the

[9]The excess score of a report that is jointly impossible with a rival signal realization receiving positive weight in (20) is set to infinity.

remainder term of Lemma 3. In other words, a report's excess score measures how much the manipulation overproduces it relative to on-path play, in the states of the world the received signal indicates are likely.

The following proposition shows that q_i is optimal if and only if any report with support under q_i minimizes $\Lambda(\cdot \mid y_i)$ when it is computed with reference to q_i. For the truth-telling case, this reduces to an easily interpretable expression: truth-telling is optimal if and only if the expected likelihood ratio of the mediator's data—the deviator's reported signal paired with rivals' true signals—is minimized by reporting the signal actually observed, where the expectation is taken with respect to i's beliefs over the actions and signals of $-i$ given its observed signal. Checking whether truth-telling is optimal therefore requires comparing $|\bar{Y}_i|^2$ numbers for each combination of $(a_i, \widetilde{a}_i)$ with positive probability.

Proposition 2. *$\chi_i^2(s_i, q_i, \alpha; p) = \underline{\chi}_i^2(s_i, \alpha)$ if and only if for all $(a_i, \widetilde{a}_i)$ arising with positive probability under (s_i, α), $y_i \in \bar{Y}_i$, and $\widetilde{y}_i$ such that $q_i(\widetilde{y}_i \mid a_i, \widetilde{a}_i, y_i) > 0$, we have*

$$\Lambda(\widetilde{y}_i \mid y_i) \leq \Lambda(\widetilde{y}'_i \mid y_i) \tag{21}$$

for all $\widetilde{y}'_i \in \bar{Y}_i$, where Λ is computed with respect to (s_i, q_i, α). In particular, $\underline{\chi}_i^2(s_i, \alpha) = \chi_i^{2,T}(s_i, \alpha)$ if and only if, for all such $(a_i, \widetilde{a}_i)$,

$$\sum_{a_{-i}, y_{-i}} \alpha(a_{-i} \mid a_i) p(y_i, y_{-i} \mid \widetilde{a}) \left(\frac{p(y_i, y_{-i} \mid \widetilde{a})}{p(y_i, y_{-i} \mid a)} - \frac{p(\widetilde{y}_i, y_{-i} \mid \widetilde{a})}{p(\widetilde{y}_i, y_{-i} \mid a)} \right) \leq 0 \tag{22}$$

for all $y_i, \widetilde{y}_i \in \bar{Y}_i$.

Proof. By Lemma 3, minimizing the detectability over q_i is equivalent to minimizing $R(\cdot)$. The components of q_i at different $(a_i, \widetilde{a}_i)$ pairs are separate variables and are additively separable in R; hence, the problem is separable across $(a_i, \widetilde{a}_i)$ pairs. Fix such a pair arising with positive probability and write $q_i(\widetilde{y}_i \mid y_i)$ for $q_i(\widetilde{y}_i \mid a_i, \widetilde{a}_i, y_i)$. Substituting $p(\widetilde{y}_i \mid y_{-i}, \widetilde{a}; q_i) = \sum_{y_i} q_i(\widetilde{y}_i \mid y_i) p(y_i \mid y_{-i}, \widetilde{a})$ into R, the problem at this pair is

$$\begin{aligned} \min_{\{q_i(\cdot \mid y_i)\}_{y_i \in \bar{Y}_i}} \quad & \sum_{a_{-i}, \widetilde{y}_i, y_{-i}} w(a, \widetilde{a}_i, (\widetilde{y}_i, y_{-i})) \left(\frac{\sum_{y_i} q_i(\widetilde{y}_i \mid y_i) p(y_i \mid y_{-i}, \widetilde{a}) - p(\widetilde{y}_i \mid y_{-i}, a)}{p(\widetilde{y}_i \mid y_{-i}, a)} \right)^2 \\ \text{s.t.} \quad & q_i(\widetilde{y}_i \mid y_i) \geq 0 \ \forall \widetilde{y}_i, y_i, \quad \sum_{\widetilde{y}_i} q_i(\widetilde{y}_i \mid y_i) = 1 \ \forall y_i, \end{aligned}$$

where w is as in the proof of Proposition 1. Let $\psi(\widetilde{y}_i \mid y_i)$ be the multiplier on the corresponding nonnegativity constraint and $\eta(y_i)$ the multiplier on y_i's simplex constraint. Differentiating the objective with respect to $q_i(\widetilde{y}_i \mid y_i)$ yields $2\alpha(a_i) s_i(\widetilde{a}_i \mid a_i) \Lambda(\widetilde{y}_i \mid y_i)$, where $\alpha(a_i) = \sum_{a_{-i}} \alpha(a)$

is the marginal probability of recommendation a_i. Absorbing the positive constant into the multipliers, the first-order condition is

$$\Lambda(\widetilde{y}_i \mid y_i) - \psi(\widetilde{y}_i \mid y_i) + \eta(y_i) = 0, \tag{23}$$

where $\psi(\widetilde{y}_i \mid y_i) \geq 0$, and $\psi(\widetilde{y}_i \mid y_i) = 0$ if $q_i(\widetilde{y}_i \mid y_i) > 0$. The objective is a convex quadratic and the constraints are linear, so these conditions are necessary and sufficient for a global minimum. If q_i is optimal and $q_i(\widetilde{y}_i \mid y_i) > 0$, subtracting (23) evaluated at $\widetilde{y}_i'$ from (23) evaluated at $\widetilde{y}_i$ gives $\Lambda(\widetilde{y}_i \mid y_i) = \Lambda(\widetilde{y}_i' \mid y_i) - \psi(\widetilde{y}_i' \mid y_i) \leq \Lambda(\widetilde{y}_i' \mid y_i)$. Conversely, if $q_i(\widetilde{y}_i \mid y_i) > 0$ implies that $\widetilde{y}_i$ minimizes $\Lambda(\cdot \mid y_i)$, then $\eta(y_i) = -\min_{\widetilde{y}_i'} \Lambda(\widetilde{y}_i' \mid y_i)$ and $\psi(\widetilde{y}_i \mid y_i) = \Lambda(\widetilde{y}_i \mid y_i) + \eta(y_i)$ satisfy (23), nonnegativity, and complementary slackness, so q_i is optimal. This establishes the first claim.

For the second claim, the support of $q_i^T(\cdot \mid y_i)$ is $\{y_i\}$, so by the first claim $\underline{\chi}_i^2 = \chi_i^{2,T}$ if and only if $\Lambda(y_i \mid y_i) \leq \Lambda(\widetilde{y}_i \mid y_i)$ for all $y_i, \widetilde{y}_i \in \bar{Y}_i$, with scores evaluated at q_i^T. At q_i^T the reported conditional is $p(\widetilde{y}_i \mid y_{-i}, \widetilde{a})$, so the term in parentheses in (20) becomes $p(\widetilde{y}_i \mid y_{-i}, \widetilde{a})/p(\widetilde{y}_i \mid y_{-i}, a) - 1$. The -1 enters both scores with identical weights and cancels from the difference, while in the remaining term the weights and the conditional likelihood ratio combine into joint distributions according to

$$\frac{p(y_{-i} \mid \widetilde{a})^2}{p(y_{-i} \mid a)} p(y_i \mid y_{-i}, \widetilde{a}) \frac{p(\widetilde{y}_i \mid y_{-i}, \widetilde{a})}{p(\widetilde{y}_i \mid y_{-i}, a)} = p(y_i, y_{-i} \mid \widetilde{a}) \frac{p(\widetilde{y}_i, y_{-i} \mid \widetilde{a})}{p(\widetilde{y}_i, y_{-i} \mid a)}.$$

Thus,

$$\Lambda(y_i \mid y_i) - \Lambda(\widetilde{y}_i \mid y_i) = \sum_{a_{-i}, y_{-i}} \alpha(a_{-i} \mid a_i) p(y_i, y_{-i} \mid \widetilde{a}) \left(\frac{p(y_i, y_{-i} \mid \widetilde{a})}{p(y_i, y_{-i} \mid a)} - \frac{p(\widetilde{y}_i, y_{-i} \mid \widetilde{a})}{p(\widetilde{y}_i, y_{-i} \mid a)} \right),$$

and requiring this difference to be nonpositive for all y_i and $\widetilde{y}_i$ is (22). □

4.2 Comparison with AK19

This subsection uses the results derived in this section to compare AK19's bounds with my bound and with SW23's bound. To state AK19's bound, we need to introduce an alternative measure of detectability. The total variation distance of rivals' signals induced by a unilateral deviation of player i to $\widetilde{a}_i$ given an action profile a is

$$TV_i(a, \widetilde{a}_i) = \frac{1}{2} \sum_{y_{-i}} |p(y_{-i} \mid a) - p(y_{-i} \mid \widetilde{a}_i, a_{-i})|.$$

Total variation distance is conceptually similar to χ^2-divergence: it measures the statistical distance between two distributions. The two are related by several standard inequalities. First, $TV_i(a, \widetilde{a}_i) \leq \frac{1}{2}\sqrt{\chi_i^2(\widetilde{a}_i, a; p_{-i})}$. Second, if $p(y \mid a) \geq \varepsilon$ for all $a \in A$ and $y \in \bar{Y}$, then $TV_i(a, \widetilde{a}_i) \geq \frac{\sqrt{\varepsilon}}{2}\sqrt{\chi_i^2(\widetilde{a}_i, a; p_{-i})}$.[10]

AK19's Proposition 4.1 states that any Nash equilibrium occupation measure α of Γ satisfies

$$\max_{i,\widetilde{a}_i} g_i(\widetilde{a}_i, \alpha) \leq \frac{2\delta}{1-\delta}\|u\|_\infty \max_{i,a,\widetilde{a}_i} TV_i(a, \widetilde{a}_i) \tag{24}$$

where $\|u\|_\infty = \sup_{i,a}|u_i(a)|$ and $g_i(\widetilde{a}_i, \alpha) = \sum_a \alpha(a)(u_i(\widetilde{a}_i, a_{-i}) - u_i(a))$. This result holding implies that α is an ε-coarse correlated equilibrium of the stage game, with ε equal to the right-hand side of (24).[11]

This result is uniform in the sense that $\max_{i,a,\widetilde{a}_i} TV_i(a, \widetilde{a}_i)$ is a statistic of the monitoring structure alone and not the putative equilibrium α. However, aspects of this uniformity that are not exploited in AK19's proof lead to several small tightenings that facilitate comparison with Theorem 1. First, the constraint on player i's gain requires only its own payoff bound $\|u_i\|_\infty = \max_a |u_i(a)|$ and the total variation of its own deviation, so the maxima over players and deviations can be paired rather than separately maximized. Second, the maximum of $TV_i(a, \widetilde{a}_i)$ over all action profiles can be replaced by its average under the equilibrium distribution itself. Finally, we can use similar techniques to Theorem 1 to extend the total variation distance under a delayed signal to an *effective total variation distance* $TV_{i,\text{eff}}(a, \widetilde{a}_i)$, where

$$TV_{i,\text{eff}}(a, \widetilde{a}_i) = (1-\delta)\sum_{\ell=0}^{L-1} \delta^\ell TV_i^{\leq \ell}(a, \widetilde{a}_i) + \delta^L\, TV_i^{\leq L}(a, \widetilde{a}_i),$$

and $TV_i^{\leq \ell}(a, \widetilde{a}_i)$ is the total variation distance between the distributions of rivals' signals through lag ℓ, $y_{-i}^{\leq \ell} \mid a$ and $y_{-i}^{\leq \ell} \mid \widetilde{a}_i, a_{-i}$. We can therefore rewrite the condition as follows:

[10] See, e.g., Polyanskiy and Wu (2025, Proposition 7.15 and Section 7.6).

[11] An ε-coarse correlated equilibrium is weaker than an ε-correlated equilibrium, the static solution concept implied by SW23 and Theorem 1, because it requires that the deviator use the same deviation $\widetilde{a}_i$ for every recommendation a_i, rather than tailoring the deviation to the conditional rival action distribution $\alpha(a_{-i} \mid a_i)$. In addition, AK19 state their result in payoff space—every Nash equilibrium payoff vector lies in the set of ε-coarse correlated equilibrium payoffs of the stage game—and with δ^2 in the numerator rather than δ, because they state the expected discounted payoffs from a path of payoffs $(u_t)_{t=1}^\infty$ as $(1-\delta)\sum_t \delta^t u_t$, whereas SW23 and this paper use $(1-\delta)\sum_t \delta^{t-1} u_t$. Equation (24) restates the statement established by their proof in this paper's conventions.

every Nash equilibrium occupation measure α satisfies

$$g_i(\widetilde{a}_i, \alpha) \leq \frac{2\delta}{1-\delta} \|u_i\|_\infty \sum_a \alpha(a)\, TV_{i,\mathrm{eff}}(a, \widetilde{a}_i) \quad \forall i = 1, \ldots, N,\ \widetilde{a}_i \in A_i. \tag{bound-AK19}$$

The formal statement and proof of this claim are in Appendix A.3 (Theorem 2).

With (bound) and (bound-AK19) both stated per player and per deviation at a common occupation measure, the two bounds can be compared directly. The first and most immediate difference is that (bound) bounds the gain from any manipulation, including those in which players deviate differently depending on what action they are "supposed" to play, while AK19 only bounds the gain from a deviation to a fixed action. This aspect of (bound) is strictly more general than (bound-AK19) and tends to matter more in games in which the best deviation depends on rivals' actions, but would matter less in games where each player has a dominant action in the stage game such as a repeated prisoner's dilemma.

However, the two bounds can be compared even when limiting to a fixed deviation. Abusing notation slightly, let $\widetilde{a}_i$ denote the constant action manipulation in which $\widetilde{a}_i$ is played with probability 1 regardless of a_i, and let $p_{\min} = \min_{a \in A, y \in \bar{Y}} p(y \mid a) > 0$, where the inequality is by the full-support assumption. Applying $TV_i(a, \widetilde{a}_i) \in \frac{1}{2}\sqrt{\chi_i^2(\widetilde{a}_i, a; p_{-i})} \times \left[\sqrt{p_{\min}}, 1\right]$, where $k \times [a, b] = [ak, bk]$, to (bound-AK19) lag-by-lag and averaging over α, the ratio of the right-hand sides of (bound-AK19) to (bound) for a fixed i, α, and $\widetilde{a}_i$ is within

$$\sqrt{\frac{\delta}{1-\delta}} \times \frac{\|u_i\|_\infty}{\sqrt{V_i(\alpha)}} \times \frac{\sum_a \alpha(a) \sum_{\ell=0}^{L} w_\ell \sqrt{\chi_i^2(\widetilde{a}_i, a; p_{-i}^{\leq \ell})}}{\sqrt{\underline{\chi}_{i,\mathrm{eff}}^2(\widetilde{a}_i, \alpha)}} \times \left[\sqrt{p_{\min}}, 1\right], \tag{25}$$

where $w_\ell = (1-\delta)\delta^\ell$ if $\ell < L$ and $w_\ell = \delta^L$. (bound) is tighter than (bound-AK19) if this ratio is greater than 1.

Each factor multiplying $[\sqrt{p_{\min}}, 1]$ in (25) isolates one structural difference between the bounds. The first reflects the rates at which discounting enters the two proofs: (bound-AK19) scales with $\delta/(1-\delta)$, while (bound) scales with its square root. This component is greater than 1 (hence points towards (bound) being tighter) whenever $\delta > 0.5$, and increases with δ, meaning (bound) becomes relatively tighter for more patient players. The second factor is at least 1; the standard deviation of $u_i(a)$ under α is at most $(\sup u_i - \inf u_i)/2$, which is at most $\|u_i\|_\infty$. This is reflective of (bound) bounding punishment with the on-path variance of payoffs, while (bound-AK19) bounds punishment with the largest possible change to payoffs, which is looser. Finally, Proposition 5—the delayed-signal counterpart to Proposition 1—along with Jensen's inequality, which implies that the numerator is at most $\sqrt{\chi_{i,\mathrm{eff}}^2(\widetilde{a}_i, \alpha; p_{-i})}$, shows that the third term is at most 1. This is what (bound-AK19) gains from using a memory-free deviation that plays $\widetilde{a}_i$ forever, and hence does not attempt to recorrelate future actions with rivals' actions, as opposed to a one-period deviation to $\widetilde{a}_i$ that then plays as though it is on path afterwards. However, the third term becomes closer to 1 if the conditions of Proposition 1(ii) are met, and hence $\chi_i^2(\widetilde{a}_i, \alpha; p_{-i}) = \underline{\chi}_i^2(\widetilde{a}_i, \alpha)$. This enables characterization of uniform circumstances in which (bound) is tighter than (bound-AK19).

Proposition 3. *Suppose that the conditions of Proposition 5(ii)—the delayed-signal counterpart to Proposition 1(ii)—hold for every i, a, and $\widetilde{a}_i$, and that $TV_i^{\leq L}(a, \widetilde{a}_i) = TV_i^{\leq L}(a', \widetilde{a}_i) = 0 \implies u_i(a) = u_i(a')$ for all i, a, a', and $\widetilde{a}_i$. Then there exists $\overline{\delta} < 1$ such that if $\delta \geq \overline{\delta}$, every α satisfying* (bound) *satisfies* (bound-AK19).

Proof. See Appendix A.4. □

Proposition 3 states that for games in which message manipulation can attain the floor and where there does not exist a fully undetectable deviation that has different payoffs for the same deviation, (bound) is uniformly tighter than (bound-AK19) for a sufficiently high discount factor.[12] Under i-semipublic monitoring, the result holds for (bound-SW23) in addition to (bound). However, in games in which players cannot fully match the on-path signal structure with a deviation, the wedge between $\chi_i^{2,T}$ and $\underline{\chi}_i^2$ precludes a uniform statement ordering (bound-SW23) and (bound-AK19).

There is no converse result—conditions under which (bound-AK19) implies (bound) uniformly—for two reasons. First, (bound) uses $V_i(\alpha)$, which approaches zero as α concentrates, instead of $\|u_i\|_\infty$, which does not depend on α. Hence, a degenerate α forces the right-hand side of (bound) to zero while that of (bound-AK19) is generally bounded away from zero whenever actions have positive detectability. Second, (bound-AK19) only constrains constant-action deviations, while (bound) constrains all deviations. At equal ε, the ε-correlated equilibrium constraints are a superset of the ε-coarse correlated ones, so (bound) can restrict the equilibrium set more even when its ε is larger.

[12] The condition $TV_i^{\leq L}(a, \widetilde{a}_i) = TV_i^{\leq L}(a', \widetilde{a}_i) = 0 \implies u_i(a) = u_i(a')$ is needed because otherwise, there would exist an action distribution that places positive weight on a and a' only with strictly positive payoff variance. Then, placing an infinitesimal amount of weight μ on a detectable profile causes (bound) to increase by $O(\sqrt{\mu})$, since it averages the detectability within a square root, but (bound-AK19) to increase by $O(\mu)$, which is smaller. Thus, such an action distribution can satisfy (bound) without satisfying (bound-AK19). Conversely, if the condition holds, then this profile would have variance approximately equal to zero and $\sqrt{V_i(\alpha)}$ would grow at rate $O(\sqrt{\mu})$, so the right-hand side of (bound) as a whole also increases at rate $O(\mu)$.

5 Application: the effect of information sharing

Using the results above, I study the limits of collusion in a repeated oligopoly, and the extent to which firms sharing information, or learning about demand ex post, affects these limits.

N symmetric firms compete by setting prices $p_i \leq \overline{p}$ in each period. Given prices, firm i's sales are log-normally distributed, with log sales

$$x_i = A - b_1 p_i + b_2 \sum_{j \neq i} p_j + \eta_i,$$

where $b_1, b_2 > 0$ and $(\eta_i)_{i=1}^N \sim \mathcal{N}(0, \Sigma)$, where Σ has unit diagonal elements and off-diagonal elements of ρ.[13] This structure implies that the vector of log sales $x \mid p \sim \mathcal{N}(\mu(p), \Sigma)$, where $p = (p_i)_{i=1}^N$ is the vector of prices and $\mu_i(p) = A - b_1 p_i + b_2 \sum_{j \neq i} p_j$. I assume $\rho \in (-\frac{1}{N-1}, 1)$, which guarantees that Σ is positive definite. Firms have constant marginal costs $c \geq 0$.

Firms attempt to maximize the discounted sum of their expected flow profits. Under these assumptions, expected flow profits conditional on p are

$$\pi_i(p) = \mathbb{E}\left[(p_i - c)\exp(x_i) \mid p\right] = (p_i - c)\exp\left(\frac{1}{2} + A - b_1 p_i + b_2 \sum_{j \neq i} p_j\right). \tag{26}$$

Below, I evaluate all flow profits in expectation, and hence drop the word "expected." I also assume that firms face a participation constraint that $p_i \geq c$ for all i. Let $\beta = b_1 - (N-1)b_2$, and maintain throughout this section the following assumption:

Assumption 1. *(i)* $b_1 > 2(N-1)b_2$ *and (ii)* $\overline{p} \in (c + 1/\beta, c + 2/b_1]$.

Condition (i) ensures that profits have an interior maximum whenever all firms set the same price, while condition (ii) ensures that profits are concave in own price over $[c, \overline{p}]$ while being above the joint profit-maximizing prices derived below.

Let $\pi(p)$ (without a subscript) be a firm's profits whenever all firms set the same price p,

[13] While the results of Sections 3 and 4 are presented with finite action and signal spaces, they all go through with infinite action and signal spaces as long as (6) has a well-defined solution, which it does throughout this section.

and let $\frac{\partial \pi(p)}{\partial p_{\text{own}}}$ and $\frac{\partial \pi(p)}{\partial p_{\text{rival}}}$ be the derivatives of flow profits at $p = (p, \ldots, p)$,

$$\begin{aligned}
\pi(p) &= (p-c)\exp\left(\frac{1}{2} + A - \beta p\right), \\
\frac{\partial \pi(p)}{\partial p_{\text{own}}} &= \exp\left(\frac{1}{2} + A - \beta p\right)(1 - b_1(p-c)), \\
\frac{\partial \pi(p)}{\partial p_{\text{rival}}} &= b_2 \pi(p).
\end{aligned} \tag{27}$$

The symmetric Nash price p^N sets $\frac{\partial \pi(p)}{\partial p_{\text{own}}}$ to zero, while the symmetric monopolist price p^M sets $\frac{\partial \pi(p)}{\partial p_{\text{own}}} + (N-1)\frac{\partial \pi(p)}{\partial p_{\text{rival}}}$ to zero. Solving for both yields the familiar log-linear markups $1/b_1$ and $1/\beta$:

$$p^N = c + \frac{1}{b_1}, \qquad \pi^N = \frac{1}{b_1}\exp\left(A - \beta c - \left(\frac{1}{2} - \frac{(N-1)b_2}{b_1}\right)\right), \tag{28}$$

$$p^M = c + \frac{1}{\beta}, \qquad \pi^M = \frac{1}{\beta}\exp\left(A - \beta c - \frac{1}{2}\right). \tag{29}$$

Our goal is to characterize the set of distributions over prices that satisfies (bound), and how that set changes given different assumptions about firms' signals. We consider three environments. First, and most minimally, firms only observe their own sales: $y_i = x_i$. Second, firms enter into an information exchange through which they ascertain the entire vector of sales: $y_i = (x_j)_{j=1}^N$.[14] Third, firms observe their own sales contemporaneously, and a market research firm publicly publishes an unbiased signal $z = \overline{\eta} + \varepsilon_z$ of the average demand shock $\overline{\eta} = N^{-1}\sum_{i=1}^N \eta_i$, where $\varepsilon_z \sim \mathcal{N}(0, \sigma_z^2)$, with a delay of L periods.

The first step in the analysis is to use the tools from Section 4 to characterize whether, and how much, firms can reduce detectability of a deviation through message manipulation.

Lemma 4. *Fix α, i, and s_i such that there exists $p \in \operatorname{supp}\alpha$ and $\widetilde{p}_i \in \operatorname{supp} s_i(\cdot \mid p_i)$ where $p_i \neq \widetilde{p}_i$. All expectations below are taken over $p \sim \alpha$ and $\widetilde{p}_i \sim s_i(\cdot \mid p_i)$.*

(a) If firms only observe their own sales,[15] then $\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i}) = \underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) < \chi^{2,T}_{i,\text{eff}}(s_i, \alpha)$, and detectability is

$$\underline{\chi}^{2,NS}_{i,\text{eff}}(s_i, \alpha) = \mathbb{E}\left[\exp\left((p_i - \widetilde{p}_i)^2 \kappa\right)\right] - 1, \tag{30}$$

where $\kappa > 0$ is the squared mean shift in x_{-i} per unit of $(p_i - \widetilde{p}_i)$, normalized by the

[14] I note that for single-product firms, this is identical to firms observing total quantities $X = \sum_j e^{x_j}$, as then each firm's rivals observe $(X, (x_j)_{j \neq i})$ and x_i can be backed out as $\log\left(X - \sum_{j \neq i} e^{x_j}\right)$.

[15] As detectability is invariant to bijective transformations, it is sufficient to consider signals as being of log sales, and I refer to sales and log sales interchangeably here and below.

inverse covariance matrix.

(b) If firms observe the full vector of sales, then $\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i}) = \underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) = \chi^{2,T}_{i,\text{eff}}(s_i, \alpha)$, *and detectability is*

$$\underline{\chi}^{2,S}_{i,\text{eff}}(s_i, \alpha) = \mathbb{E}\left[\exp\left((p_i - \widetilde{p}_i)^2\left(\kappa + \frac{\Delta^2}{\sigma^2_{i|-i}}\right)\right)\right] - 1, \tag{31}$$

where $\Delta > 0$ *is the conditional mean shift in* x_i *given* x_{-i}, p *per unit of* $(p_i - \widetilde{p}_i)$, *and* $\sigma^2_{i|-i}$ *is the conditional variance of* $x_i \mid x_{-i}, p$.

(c) If firms observe their own sales plus the signal z *with a delay of* L *periods, then* $\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i}) \leq \underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) < \chi^{2,T}_{i,\text{eff}}(s_i, \alpha)$, *where the first inequality holds with equality if and only if* $L = 0$, *and detectability is*

$$\begin{aligned}\underline{\chi}^{2,z(L)}_{i,\text{eff}}(s_i, \alpha) = (1 - \delta^L)\mathbb{E}&\left[\exp\left((p_i - \widetilde{p}_i)^2\left(\kappa + \frac{(\Delta - \Delta^*(p_i, \widetilde{p}_i))^2}{\sigma^2_{i|-i}}\right)\right)\right] \\ &+ \delta^L\mathbb{E}\left[\exp\left((p_i - \widetilde{p}_i)^2\left(\kappa_z + \frac{(\Delta_z - \Delta^*(p_i, \widetilde{p}_i))^2}{\sigma^2_{i|-i,z}}\right)\right)\right] - 1,\end{aligned} \tag{32}$$

where $\kappa_z > 0$ *is the normalized squared mean shift in* x_{-i} *per unit of* $(p_i - \widetilde{p}_i)$ *conditional on* z, $\Delta_z \in (0, \Delta)$ *is the conditional mean shift in* x_i *given* x_{-i}, p, z *per unit of* $(p_i - \widetilde{p}_i)$, $\sigma^2_{i|-i,z}$ *is the conditional variance of* $x_i \mid x_{-i}, p, z$, *and* $\Delta^*(p_i, \widetilde{p}_i) \in [\Delta_z, \Delta]$ *minimizes* (32) *for each* $(p_i, \widetilde{p}_i)$.

The precise definition of all variables and derivation of these formulas are in Appendix A.5. They have the following intuition. Using the chain rule for χ^2-divergences (alternatively, Lemma 3), we can rewrite the detectabilities using the detectability from x_{-i} alone, and x_i conditional on x_{-i}. Due to the linear-Gaussian demand structure, price deviations only affect expected log sales without affecting their variance. In case (a), a message manipulation that "de-biases" this mean shift attains the rivals-only detectability floor.[16] Case (b) is public monitoring, so Corollary 1 applies, and the detectability is that of the full sales vector. Finally, in case (c), the deviator can condition its report on the public signal and still reduce the detectability of its own signal to zero when $L = 0$. However, when $L > 0$, the deviator cannot simultaneously zero out the pre-signal and post-signal detectability. Using the sufficient condition of Proposition 2, we show that a deterministic manipulation that reports $\widetilde{x}_i = x_i - \Delta^*(p_i, \widetilde{p}_i)(p_i - \widetilde{p}_i)$, where the per-unit recentering $\Delta^*(p_i, \widetilde{p}_i)$ is chosen to balance the pre-signal and post-signal detectability appropriately, is globally optimal.

[16] One can also show this by rewriting the signals so that Corollary 1(ii) is satisfied.

With these formulas, we can directly characterize the set of α satisfying (bound). Define firm i's *interim profits* as the expected profits from setting price $\widetilde{p}_i$ when firm i is recommended price p_i,

$$\pi_i(\widetilde{p}_i \mid p_i; \alpha) = \mathbb{E}_\alpha \left[\pi_i(\widetilde{p}_i, p_{-i}) \mid p_i\right] = (\widetilde{p}_i - c) \exp\left(\frac{1}{2} + A - b_1 \widetilde{p}_i\right) \mathbb{E}_\alpha \left[\exp\left(b_2 \sum_{j \neq i} p_j\right) \middle| p_i\right],$$

and the *interim profit slope* as the derivative of interim profits, evaluated at $\widetilde{p}_i = p_i$,

$$\begin{aligned} D_i(p_i; \alpha) &= \frac{\partial}{\partial \widetilde{p}} \mathbb{E}_\alpha \left[\pi_i(\widetilde{p}_i, p_{-i}) \mid p_i\right] \Big|_{\widetilde{p}_i = p_i} \\ &= (1 - b_1(p_i - c)) \exp\left(\frac{1}{2} + A - b_1 p_i\right) \mathbb{E}_\alpha \left[\exp\left(b_2 \sum_{j \neq i} p_j\right) \middle| p_i\right]. \end{aligned} \tag{33}$$

The interim profit slope is the marginal expected gain from an infinitesimal price deviation given one's own recommendation. Due to the log-linear structure, interim profits factor multiplicatively and are concave in $\widetilde{p}_i$ for every α. This implies that a necessary and sufficient condition for a deviation to be profitable (in the sense of the bound) is that an infinitesimal deviation in the same direction is also profitable. We can therefore prove that (bound) collapses to a single moment condition relating the gain from an infinitesimal deviation to the change to detectability implied by that deviation.

Proposition 4. *Under each monitoring structure $o \in \{NS, S, z(L)\}$ (no sharing, sharing sales, and public demand signal with lag L respectively), there exists a* monitoring constant *K_o such that any distribution α satisfies* (bound) *if and only if, for every i,*

$$\mathbb{E}_{\alpha_i} \left[D_i(p_i; \alpha)^2\right] \leq \frac{\delta}{1 - \delta} K_o^2 V_i(\alpha), \tag{34}$$

where α_i is the marginal of p_i over α and the monitoring constants satisfy $K_{NS} = \sqrt{\kappa}$, $K_S = \sqrt{\kappa + \Delta^2/\sigma^2_{i|-i}}$, and

$$K_{z(L)} = \sqrt{(1 - \delta^L)\kappa + \delta^L \kappa_z + \frac{\delta^L (1 - \delta^L)(\Delta - \Delta_z)^2}{(1 - \delta^L)\sigma^2_{i|-i,z} + \delta^L \sigma^2_{i|-i}}}. \tag{35}$$

Proof. See Appendix A.6. □

The limits of collusion under a particular monitoring structure thereby boil down to a single constant, K_o. The constant has several interpretations. Perhaps the most intuitive is that it is the smallest possible change to the mean signal distribution—in Mahalanobis distance—per unit of price deviation. It can also be interpreted as the Fisher information transmitted by the signal. For a Gaussian mean-shift family, the χ^2-divergence implied by a mean shift of h is $\exp(h^2/\sigma^2) - 1$, where $1/\sigma^2$ is the Fisher information, while here, the detectability of a price shift of Δp is $\exp(\Delta p^2 K_o^2) - 1$ (exactly so under NS and S; under $z(L)$, a δ^L-mixture of two such terms).[17]

The particular functional form of (34) also enables direct comparison of different monitoring structures. Because K_o^2 multiplies $\delta/(1-\delta)$, the effect of moving from one monitoring structure to another is isomorphic to the effect of changing firms' discount factors.

Corollary 2. *The following are equivalent:*

(i) α satisfies (bound) *under no information sharing at discount factor δ.*

(ii) α satisfies (bound) *under monitoring structure o and discount factor δ_o solving*

$$\frac{\delta}{1-\delta} = \frac{K_o^2}{K_{NS}^2} \cdot \frac{\delta_o}{1-\delta_o}.$$

For example, suppose $b_1 = 9b_2$, $\rho = 0.6$, $N = 5$, and $\delta = 0.9$ (so $\delta/(1-\delta) = 9$). These parameters imply that $K_S^2/K_{NS}^2 \approx 141$. Therefore, the effect of information sharing on the set of distributions satisfying (bound) is the same as the effect of increasing $\delta/(1-\delta)$ to approximately 1269, and hence δ to approximately 0.9992.

This direct comparison admits a clean ordering between the cases: throughout the parameter space, we have $K_{NS} < K_{z(L)} < K_S$. The first inequality is clear from visual inspection of K_{NS} and $K_{z(L)}$, along with the fact that $\kappa_z > \kappa$. As $\sigma_z^2 \to \infty$, κ_z approaches κ, while as $L \to \infty$ or $\delta \to 0$, δ^L approaches zero; in either case, $K_{z(L)}$ approaches K_{NS} in the limit. As for the second inequality, direct computation gives that $K_S > K_{z(L)}$ if and only if

$$b_1 + \frac{\rho(N-1)b_2}{1+\rho(N-2)} > \frac{(N-1)b_2(1+\rho(N-1))}{1+\rho(N-2)} \times \sqrt{\frac{\delta^L(1-\rho)(1+\rho(N-1))}{N^2\sigma_z^2(1+\rho(N-2)) + \delta^L(1-\rho)(1+\rho(N-1))}}.$$

The term inside the square root is at most one, so a sufficient condition is that the left-hand side is greater than the first term on the right-hand side. This collapses to $b_1 > (N-1)b_2$, which is guaranteed by Assumption 1. When b_1 is high enough that own-price effects always dominate cross-price effects, knowing a firm's sales always provides more information about that firm's price than knowing aggregate demand conditions, conditional on knowing rivals' sales.

[17] For a signal structure with lags, K_o can be interpreted as the smallest possible weighted average of the Fisher information of the signals using the same weights as effective detectability.

5.1 Comparative statics

This subsection studies the comparative statics of primitives on two types of quantities. First, we may be interested in how primitives affect the set of equilibria within monitoring structure o. Proposition 4 shows that this is isomorphic to studying the partial derivative of the monitoring constant K_o with respect to primitives. Second, we may be interested in how primitives affect the impact of moving from monitoring structure o to monitoring structure o'. Corollary 2 shows that this is isomorphic to studying the partial derivative of the ratio of the two monitoring constants, $R_{o,o'} \equiv K_{o'}/K_o$, with respect to primitives.

No information sharing. K_{NS} is unaffected by the own-price semi-elasticity b_1, as changes to b_1 do not affect the extent to which rivals can detect deviations using only their sales. It is increasing in b_2, the cross-price semi-elasticity, for the complementary reason: it directly affects the extent to which price changes are reflected in rivals' sales. It is weakly decreasing in ρ (strictly if $N > 2$, and exactly zero if $N = 2$) and increasing in N because the main inference problem is disentangling price deviations from demand shocks. With highly correlated demand, or with relatively few firms, a shock that reduces all firms' sales by as much as a rivals' price deviation is relatively plausible, and becomes less plausible as demand becomes more uncorrelated and as the number of firms increases.

Sharing sales. K_S is clearly increasing in b_1, and is also increasing in N for similar reasons to K_{NS}. However, perhaps surprisingly, it is not monotone in b_2 or ρ: the signs of $\partial K_S/\partial b_2$ and $\partial K_S/\partial \rho$ are the same as the sign of $\rho b_1 + b_2$. Thus, along both metrics, detectability is minimized when $\rho b_1 + b_2 = 0$, and increases if ρ or b_2 change in either direction from this point. To understand the intuition for why, consider that a one-unit price cut for firm i increases firm i's sales by b_1 and decreases all other firms' sales by b_2, while if firm i's sales increase by b_1 at the recommended price, then the conditional mean of rivals' sales is decreased by b_2 exactly when $\rho = -b_2/b_1$. Therefore, the change to sales from a price deviation and a change to sales from a demand shock "look like" one another, minimizing detectability.

We next analyze the effect of information sharing, $R_{NS,S}$. This object is increasing in b_1 and decreasing in b_2, so industries with greater own-price elasticities and lesser cross-price elasticities stand to gain more from information sharing. $R_{NS,S}$ is, however, ambiguous in ρ and N. In ρ, there again exists a $\rho^*_{NS,S}$ such that $\operatorname{sign} \partial R_{NS,S}/\partial \rho = \operatorname{sign} \rho - \rho^*_{NS,S}$, where

$$\rho^*_{NS,S} = -\frac{b_1(N-2) + 2b_2(N-1)}{b_1((N-1)^2+1) + b_2(N-1)(N-2)}.$$

$\rho^*_{NS,S}$ is weakly lower than $-b_2/b_1$, the local minimum of K_S (strictly if $N > 2$), but greater than $-1/(N-1)$. Sharing is at its most valuable when full sales vectors are nearly degenerate conditional on prices—$\rho \downarrow -1/(N-1)$ or $\rho \uparrow 1$—and hence a deviation can be perfectly detected from rivals' sales alone. When errors are approximately independent, adding a deviator's sales does not add much more information about whether they deviated on top of adding its rivals' sales.

Lagged demand signal. $K_{z(L)}$ is increasing in b_2 and invariant to b_1 for similar reasons as K_{NS}. It is decreasing in σ_z^2, as a less informative signal provides less information, and in L, as a later-arriving signal provides less deterring power. Meanwhile, the comparative statics with respect to ρ are governed by the *effective noise* of the signal—the ratio σ_z^2/δ^L. This quantity reflects the fact that reductions in signal delay are isomorphic to increases in the precision of the signal in the magnitude of $K_{z(L)}$.[18] If the effective noise of the signal is less than $1/(N-2)$, then $K_{z(L)}$ reaches a minimum in ρ whenever $\frac{1+(N-1)\rho}{N} = (N-2)\sigma_z^2/\delta^L$. Here, the left-hand side is the variance of $\overline{\eta}$, while the right-hand side is effective noise level of the signal, adjusted by the market size (where a higher market size reduces noise in $\overline{\eta}$); when these two forces are equated, the demand signal adds the least possible amount of information. If ρ increases, then the underlying noise in demand increases and the signal becomes relatively more informative, while if ρ decreases, then learning about $\overline{\eta}$ provides less information about deviations regardless. If $N = 2$ then $K_{z(L)}$ is always increasing in ρ, as the latter channel is not present with only one rival, while if $\sigma_z^2/\delta^L > 1/(N-2)$, then $K_{z(L)}$ is always decreasing in ρ, as higher correlation makes a deviation harder to distinguish from a common shock.

Finally, we characterize the comparative statics of $R_{NS,z(L)}$. This quantity is independent of b_1 and b_2 and, perhaps obviously, is decreasing in σ_z^2 and L. It is, however, increasing in ρ, even as $K_{z(L)}$ is nonmonotone in ρ: as shocks become more positively correlated, aggregate demand becomes increasingly important, while rivals' sales become more redundant as separate signals. Finally, $R_{NS,z(L)}$ is ambiguous in N. Adding competitors directly improves no-sharing monitoring while also raising the effective precision of the aggregate demand signal. Which force dominates depends on the correlation of demand and the effective noise of the signal.

Overall, there are several predictions from these comparative statics. First, without observing rivals' sales, collusion is least feasible in industries with highly correlated demand and low cross-price elasticities. Sharing verified sales information is most valuable in settings

[18] To see this, note that by substituting primitives into $K_{z(L)}$ we can rewrite it as

$$K_{z(L)}^2 = b_2^2(N-1)N\frac{N\sigma_z^2/\delta^L + (N-1)\rho + 1}{N^2(1+(N-2)\rho)\sigma_z^2/\delta^L + (1-\rho)(1+(N-1)\rho)}$$

where σ_z^2, δ, and L enter only as the ratio σ_z^2/δ^L.

where firms face relatively elastic demand, relatively differentiated products, and relatively correlated demand shocks, of either sign—when shocks are roughly independent, rivals' sales already reveal most of what the deviator's own sales would add. Finally, signals of aggregate demand are most valuable in facilitating outcomes above what can be sustained without information sharing when demand is correlated. These predict that information exchanges may take different forms in industries with different underlying fundamentals. Industries with correlated demand may find that public demand information suffices to facilitate collusion while being relatively less risky in a legal sense, while industries with relatively uncorrelated demand may find that only sharing direct sales information provides sufficient monitoring power to facilitate collusion.

6 Conclusion

This article studies bounds on outcomes in repeated games with arbitrary monitoring structures. It sharpens the result of Sugaya and Wolitzky (2023) in settings with private information, which is arguably the case of interest when studying private monitoring games. It also extends their result to settings in which the signal is revealed over multiple future periods. It demonstrates necessary and sufficient conditions for this result to constitute a strict tightening over their result and shows that this result provides a tighter bound than that of Awaya and Krishna (2019) in the patient limit—a result that is generically unavailable using the original bound of Sugaya and Wolitzky (2023). Finally, using these results, it studies the limits of collusion in a repeated oligopoly. It demonstrates that the set of equilibria can be boiled down to a single statistic—the monitoring constant—which reflects the rate at which signal distributions are changed per unit of price change, and can be traded off one-for-one with discounting. Comparative statics of this object, and those of its ratio across monitoring structures, provide testable implications for how the structure of information exchanges relates to primitives.

Ultimately, these results—and the literature they contribute to—provide new methods of studying outcomes in repeated games with private monitoring. While such games are economically important, outcomes of these games are known to be far more difficult to predict using standard equilibrium analysis than those of games with complete information or public monitoring structures. These bounds provide a credible way of studying private monitoring games, even when the exact set of equilibria cannot be characterized.

A Additional results and proofs

A.1 Results of Section 4 with delayed signals

Propositions 5 and 6 extend Propositions 1 and 2 to the delayed-signal case. The extension of Proposition 1 is relatively straightforward, although it is worth noting that unlike the no-delay case, the distribution of $y_i^{\leq \ell} \mid \widetilde{a}_i, a_{-i}, y_{-i}^{\leq \ell}$ Blackwell-dominating the distribution of $y_i^{\leq \ell} \mid a_i, a_{-i}, y_{-i}^{\leq \ell}$ for all ℓ is, by itself, insufficient for $\underline{\chi}^2_{i,\text{eff}} = \chi^2_{i,\text{eff}}(\cdot; p_{-i})$; a slightly stronger condition is needed, because we in addition require that the garbling kernels at different values of ℓ comprise one internally consistent message manipulation, such that marginalizing $q_i^{\leq \ell}$ over $\widetilde{y}_i^{\ell}$ recovers $q_i^{\leq \ell - 1}$. Intuitively, since the mediator remembers the past reports, the deviator not only needs to perfectly replicate the distributions of y_i^{ℓ} and $y_i^{\ell-1}$, but also needs to have the latter message manipulation be consistent with the former.

As for Proposition 2, two features of the delayed-signal case require care. First, the objects entering effective detectability are the cumulative kernels $q_i^{\leq \ell}$, which share the underlying level kernels across levels; the relevant convexity is in the cumulative kernels themselves, and it holds because randomizing once, before the first report, over which of two manipulations to follow mixes every cumulative kernel linearly. Second, because the level-ℓ excess score attaches to the entire report path through lag ℓ, the optimal report at a given lag must weigh not only its contemporaneous score but also the scores of the report paths it makes reachable at later lags. The optimality condition therefore takes the form of a variational inequality—equivalently, of truth-telling solving a dynamic program—rather than a report-by-report support condition. For $\ell < L$, let $w_\ell = (1-\delta)\delta^{\ell}$, and let $w_L = \delta^L$, so that the w_ℓ are the weights in (4).

Proposition 5. *Let $\bar{Y}_{-i}^{\leq \ell}$ be the set of possible realizations of $y_{-i}^{\leq \ell}$ from $\bar{Y}$ and, for a given manipulation q_i, let $q_i^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid a_i, \widetilde{a}_i, y_i^{\leq \ell}) = \prod_{\ell'=0}^{\ell} q_i^{\ell'}(\widetilde{y}_i^{\ell'} \mid a_i, \widetilde{a}_i, y_i^0, \ldots, y_i^{\ell'}, \widetilde{y}_i^0, \ldots, \widetilde{y}_i^{\ell'-1})$. For all α and s_i,*

$$\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i}) \leq \underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) \leq \chi^{2,T}_{i,\text{eff}}(s_i, \alpha). \tag{36}$$

In addition,

(i) $\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i}) = \chi^{2,T}_{i,\text{eff}}(s_i, \alpha)$—and hence both are equal to $\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha)$—if and only if for all $a \in \operatorname{supp} \alpha$, $\widetilde{a}_i \in \operatorname{supp} s_i(\cdot \mid a_i)$, $\ell = 0, \ldots, L$, and $y_{-i}^{\leq \ell} \in \bar{Y}_{-i}^{\leq \ell}$, $p(y_i^{\leq \ell} \mid a, y_{-i}^{\leq \ell}) = p(y_i^{\leq \ell} \mid \widetilde{a}_i, a_{-i}, y_{-i}^{\leq \ell})$;

(ii) $\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) = \chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i})$ if and only if there exists a message manipulation q_i such that for all $a \in \operatorname{supp} \alpha$, $\widetilde{a}_i \in \operatorname{supp} s_i(\cdot \mid a_i)$, $\ell = 0, \ldots, L$, and $y_{-i}^{\leq \ell} \in \bar{Y}_{-i}^{\leq \ell}$, $p(\cdot \mid a, y_{-i}^{\leq \ell})$ is

a garbling of $p(\cdot \mid \widetilde{a}_i, a_{-i}, y_{-i}^{\leq \ell})$ with q_i as the Markov kernel; that is,

$$p(\widetilde{y}_i^{\leq \ell} \mid a, y_{-i}^{\leq \ell}) = \sum_{y_i^{\leq \ell}} q_i^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid a_i, \widetilde{a}_i, y_i^{\leq \ell})\, p(y_i^{\leq \ell} \mid \widetilde{a}_i, a_{-i}, y_{-i}^{\leq \ell}).$$

Proof. Effective detectability (4) is a sum with strictly positive weights over cumulative levels $\ell = 0, \ldots, L$ of the detectabilities $\chi_i^2(s_i, q_i^{\leq \ell}, \alpha; p^{\leq \ell})$, and Lemma 3 applies at each level with $(\widetilde{y}_i^{\leq \ell}, y_{-i}^{\leq \ell})$ in place of $(\widetilde{y}_i, y_{-i})$. Thus, level-$\ell$ detectability can be decomposed into a rivals-only term plus a nonnegative remainder $R^{\leq \ell}$ that depends on q_i only through $q_i^{\leq \ell}$. Summing across levels with the weights delivers the first inequality in (36), with $\chi^2_{i,\text{eff}}(s_i, \alpha; p_{-i})$ the corresponding weighted combination of the rivals-only terms. The second inequality is immediate from the definition of $\underline{\chi}^2_{i,\text{eff}}$ (6).

Because the weights are positive, a manipulation attains the lower bound if and only if $R^{\leq \ell} = 0$ for all ℓ. For (i), the argument of Proposition 1(i) applied at each level shows that the truthful manipulation satisfies $R^{\leq \ell} = 0$ for all ℓ if and only if the stated condition holds at every level. For (ii), if the stated manipulation exists, using it sets $R^{\leq \ell} = 0$ for all ℓ, attaining the lower bound. Conversely, if the lower bound is attained, it is attained by some manipulation q_i; $R^{\leq \ell} = 0$ at every level then states precisely that the cumulative kernels of this q_i satisfy the garbling condition, so the minimizer itself is the required manipulation. □

Proposition 6. *$\chi^2_{i,\text{eff}}(s_i, q_i, \alpha) = \underline{\chi}^2_{i,\text{eff}}(s_i, \alpha)$ if and only if, for every $(a_i, \widetilde{a}_i)$ arising with positive probability under (s_i, α) and every message manipulation q_i',*

$$\sum_{\ell=0}^{L} w_\ell \sum_{y_i^{\leq \ell}, \widetilde{y}_i^{\leq \ell}} \Big(q_i'^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell}) - q_i^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell}) \Big) \Lambda^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell}) \geq 0, \tag{37}$$

where $\Lambda^{\leq \ell}$, the lag-ℓ excess score, *is defined as*

$$\Lambda^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell}) = \sum_{a_{-i}, y_{-i}^{\leq \ell}} \alpha(a_{-i} \mid a_i) \frac{p(y_{-i}^{\leq \ell} \mid \widetilde{a})^2}{p(y_{-i}^{\leq \ell} \mid a)} p(y_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, \widetilde{a}) \left(\frac{p(\widetilde{y}_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, \widetilde{a}; q_i) - p(\widetilde{y}_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, a)}{p(\widetilde{y}_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, a)} \right),$$

with $p(\widetilde{y}_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, \widetilde{a}; q_i) = \sum_{y_i^{\leq \ell}} q_i^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid a_i, \widetilde{a}_i, y_i^{\leq \ell}) p(y_i^{\leq \ell} \mid y_{-i}^{\leq \ell}, \widetilde{a})$. In particular, $\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) = \chi^{2,T}_{i,\text{eff}}(s_i, \alpha)$ if and only if, with scores evaluated at q_i^T, the truthful reporting policy minimizes $\sum_\ell w_\ell \sum_{y_i^{\leq \ell}, \widetilde{y}_i^{\leq \ell}} q_i'^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell}) \Lambda^{\leq \ell}(\widetilde{y}_i^{\leq \ell} \mid y_i^{\leq \ell})$ over message manipulations q_i'.

Proof. Fix a pair $(a_i, \widetilde{a}_i)$ arising with positive probability; as in the proof of Proposition 2, the problem separates across such pairs. By Lemma 3 applied at each level, minimizing effective detectability over q_i is equivalent to minimizing $\sum_\ell w_\ell R^{\leq \ell}$, and $R^{\leq \ell}$ depends on the manipulation only through the cumulative kernel $q_i^{\le \ell}$, in which it is a convex quadratic: the cumulative reported conditional is linear in $q_i^{\le \ell}$. The set of attainable cumulative kernel sequences $(q_i^{\le \ell})_{\ell=0}^{L}$ is convex. To see this, given manipulations q_i, q_i' and $\lambda \in [0,1]$, define $m^{\le \ell} = \lambda q_i^{\le \ell} + (1-\lambda) q_i'^{\le \ell}$ and level kernels $q_i''^{\ell} = m^{\le \ell}/m^{\le \ell - 1}$ (defined arbitrarily when the denominator vanishes). Each such kernel's rows sum to one because marginalizing a cumulative kernel over $\widetilde{y}_i^{\ell}$ recovers its predecessor, and the cumulative kernels of q_i'' equal exactly $m^{\le \ell}$ at each ℓ, including along zero-probability report paths. The objective is therefore convex on a convex set, and q_i is a global minimizer if and only if no feasible direction has a strictly negative directional derivative. Differentiating $R^{\le \ell}$ with respect to the entries of $q_i^{\le \ell}$ yields $2\alpha(a_i) s_i(\widetilde{a}_i \mid a_i) \Lambda^{\le \ell}$, exactly as in the proof of Proposition 2 with paths in place of signals and reports; absorbing the positive constant and summing across levels, the directional derivative toward q_i' is proportional to the left side of (37), which gives the first claim.

For the second claim, evaluate (37) at $q_i = q_i^T$. Since the q_i^T term is a constant, the condition states that q_i^T minimizes the linear functional $q_i' \mapsto \sum_\ell w_\ell \sum q_i'^{\le \ell} \Lambda^{\le \ell}$ over message manipulations. This minimization is a finite-horizon dynamic program with the stated flow costs, solvable by backward induction over report histories. When $L = 0$ there is a single level and the program reduces to minimizing $\Lambda(\cdot \mid y_i)$ separately at each received signal, recovering Proposition 2. □

A.2 Proof of Lemma 3

Fix a recommended action profile a and a manipulated action profile $\widetilde{a} = (\widetilde{a}_i, a_{-i})$. For brevity, write $p(\widetilde{y}) = p(\widetilde{y} \mid a)$ and $\widetilde{p}(\widetilde{y}) = p(\widetilde{y} \mid \widetilde{a}; q_i)$ for the on-path and manipulated distributions of the report profile $\widetilde{y} = (\widetilde{y}_i, y_{-i})$. Recall that for any two distributions $P(x)$ and $Q(x)$, $1 + \chi^2(Q \| P) = \sum_x Q(x)^2 / P(x)$. Therefore,

$$
\begin{aligned}
1 + \sum_{\widetilde{y}} p(\widetilde{y}) \left(\frac{\widetilde{p}(\widetilde{y}) - p(\widetilde{y})}{p(\widetilde{y})} \right)^2 &= \sum_{\widetilde{y}} \frac{\widetilde{p}(\widetilde{y})^2}{p(\widetilde{y})} \\
&= \sum_{y_{-i}} \frac{\widetilde{p}(y_{-i})^2}{p(y_{-i})} \sum_{\widetilde{y}_i} \frac{\widetilde{p}(\widetilde{y}_i \mid y_{-i})^2}{p(\widetilde{y}_i \mid y_{-i})} \\
&= \sum_{y_{-i}} \frac{\widetilde{p}(y_{-i})^2}{p(y_{-i})} \left(1 + \sum_{\widetilde{y}_i} p(\widetilde{y}_i \mid y_{-i}) \left(\frac{\widetilde{p}(\widetilde{y}_i \mid y_{-i}) - p(\widetilde{y}_i \mid y_{-i})}{p(\widetilde{y}_i \mid y_{-i})} \right)^2 \right) \\
&= 1 + \sum_{y_{-i}} p(y_{-i}) \left(\frac{\widetilde{p}(y_{-i}) - p(y_{-i})}{p(y_{-i})} \right)^2 + \\
&\qquad \sum_{y_{-i}} \frac{\widetilde{p}(y_{-i})^2}{p(y_{-i})} \sum_{\widetilde{y}_i} p(\widetilde{y}_i \mid y_{-i}) \left(\frac{\widetilde{p}(\widetilde{y}_i \mid y_{-i}) - p(\widetilde{y}_i \mid y_{-i})}{p(\widetilde{y}_i \mid y_{-i})} \right)^2 .
\end{aligned}
$$

Subtracting 1 from both sides and taking expectations against α and s_i yields the result. $\square$

A.3 Extending AK19 to delayed signals

This appendix states and proves the tightened form of AK19's bound given in Section 4.2, extended to delayed monitoring structures.

For probability distributions P and Q on a common finite set, write $\|P - Q\|_{TV} = \frac{1}{2}\sum_x |P(x) - Q(x)|$ for their total variation distance. For $\ell \in \{0, \ldots, L\}$, let $p_{-i}^{\leq \ell}(\cdot \mid a)$ denote the marginal distribution of $p(\cdot \mid a)$ over rivals' signal components through lag ℓ, $y_{-i}^{\leq \ell} = (y_{-i}^{\ell'})_{\ell'=0}^{\ell}$. Extending the notation of Section 4.2, define the cumulative footprint of a deviation on rivals' signals, and its average under a distribution $\alpha \in \Delta(A)$, as

$$TV_i^{\leq \ell}(a, \widetilde{a}_i) = \left\| p_{-i}^{\leq \ell}(\cdot \mid a) - p_{-i}^{\leq \ell}(\cdot \mid \widetilde{a}_i, a_{-i}) \right\|_{TV}$$
$$TV_i^{\leq \ell}(\widetilde{a}_i, \alpha) = \sum_a \alpha(a)\, TV_i^{\leq \ell}(a, \widetilde{a}_i).$$

The *effective total variation distance* induced by deviation $\widetilde{a}_i$ against α is the discounted combination of cumulative distances

$$TV_{i,\text{eff}}(\widetilde{a}_i, \alpha) = (1-\delta)\sum_{\ell=0}^{L-1} \delta^\ell TV_i^{\leq \ell}(\widetilde{a}_i, \alpha) + \delta^L TV_i^{\leq L}(\widetilde{a}_i, \alpha), \tag{38}$$

which has the same form as the effective detectability (4). When $\mathcal{L} = \{0\}$, $TV_{i,\text{eff}}(\widetilde{a}_i, \alpha) = \sum_a \alpha(a)\, TV_i(a, \widetilde{a}_i)$.

Theorem 2. *Every Nash equilibrium occupation measure α of Γ satisfies, for each player i and action $\widetilde{a}_i \in A_i$,*

$$g_i(\widetilde{a}_i, \alpha) \leq \frac{2\delta}{1-\delta}\, \|u_i\|_\infty\, TV_{i,\text{eff}}(\widetilde{a}_i, \alpha), \tag{39}$$

and is thus a static ε-coarse correlated equilibrium action distribution for

$$\varepsilon = \max_{i, \widetilde{a}_i} \frac{2\delta}{1-\delta} \|u_i\|_\infty TV_{i,\text{eff}}(\widetilde{a}_i, \alpha).$$

Proof. We start with some notation. A *history* for player i at the end of period t collects everything the player has observed through period t,

$$h_i^t = \left(a_{i,t'}, (y_{i,t'-\ell}^{\ell})_{\ell=0}^{\min\{t'-1, L\}} \right)_{t'=1}^{t}.$$

A *rivals' history* collects the histories of all players except for i, $h^t_{-i} = (h^t_j)_{j\neq i}$. Strategies $\sigma_i(\cdot \mid h^t_i)$ are distributions over actions $a_{i,t} \in A_i$, and rivals' period-$(t+1)$ actions are drawn from $\sigma_{-i}(\cdot \mid h^t_{-i}) = \prod_{j\neq i}\sigma_j(\cdot \mid h^t_j)$. Under a strategy profile $\sigma = (\sigma_i)_{i\in I}$, let α_t be the equilibrium distribution of period-t actions, and let $\alpha = (1-\delta)\sum_t \delta^{t-1}\alpha_t$ be the equilibrium occupation measure. Finally, for a given player i, let $\widetilde{\sigma}_i$ denote the permanent deviation that plays $\widetilde{a}_i$ in every period after every history, and let μ^t_{-i} and $\widetilde{\mu}^t_{-i}$ denote the distributions of h^t_{-i} under σ and under $(\widetilde{\sigma}_i, \sigma_{-i})$ respectively.

Lemma 5. *For every strategy profile σ, player i, action $\widetilde{a}_i$, and period $t \geq 1$,*

$$\left\|\widetilde{\mu}^t_{-i} - \mu^t_{-i}\right\|_{TV} \leq \sum_{t'=1}^{t} TV_i^{\leq \min\{t-t',L\}}(\widetilde{a}_i, \alpha_{t'}).$$

Proof. For $t' = 0, 1, \ldots, t$, let $\widetilde{\sigma}^{t'}$ denote the hybrid strategy profile in which player i follows σ_i in periods $1, \ldots, t'$ and plays $\widetilde{a}_i$ thereafter, while players $-i$ follow σ_{-i} throughout. Let $\pi^{t'}_{-i}$ denote the distribution of h^t_{-i} under $\widetilde{\sigma}^{t'}$. Then $\pi^t_{-i} = \mu^t_{-i}$ and $\pi^0_{-i} = \widetilde{\mu}^t_{-i}$. By the triangle inequality,

$$\left\|\widetilde{\mu}^t_{-i} - \mu^t_{-i}\right\|_{TV} \leq \sum_{t'=1}^{t} \left\|\pi^{t'-1}_{-i} - \pi^{t'}_{-i}\right\|_{TV}. \tag{40}$$

Fix $t' \geq 1$. Conditional on a history, the hybrids $\widetilde{\sigma}^{t'-1}$ and $\widetilde{\sigma}^{t'}$ prescribe identical play in every period except period t': both follow σ exactly through period $t'-1$; in period t', player i draws its action from σ_i under $\widetilde{\sigma}^{t'}$ but plays $\widetilde{a}_i$ under $\widetilde{\sigma}^{t'-1}$; and in periods $t'+1, \ldots, t$, player i plays the constant $\widetilde{a}_i$ under both, regardless of history.

To turn potentially random mixed actions and signals into deterministic functions, we construct the random variables generating this randomness on a common probability space explicitly. Let $(U^a_{j,\tau})_{j\in I, \tau\geq 1}$ and $(U^y_\tau)_{\tau\geq 1}$ be mutually independent random variables each distributed uniformly on $[0,1]$. We interpret $U^a_{j,\tau}$ as player j's period-τ *action seed*, and U^y_τ as the common period-τ *signal seed*. Letting Σ_j be the CDF of players' strategies σ_j, and P the CDF of p, define functions $(f^a_{j,\tau})_{j\in I,\tau\geq 1}$ and $(f^y_\tau)_{\tau\geq 1}$ as

$$\begin{aligned} f^a_{j,\tau}(h^{\tau-1}_j, U^a_{j,\tau}) &= \Sigma_j^{-1}(U^a_{j,\tau} \mid h^{\tau-1}_j), \\ f^y_\tau(a_\tau, U^y_\tau) &= P^{-1}(U^y_\tau \mid a_\tau). \end{aligned}$$

Given that $U^a_{j,\tau}$ and U^y_τ are uniformly distributed, these constructions guarantee that $f^a_{j,\tau}(h^{\tau-1}_j, U^a_{j,\tau})$ is distributed according to $\sigma_j(\cdot \mid h^{\tau-1}_j)$ and $f^y_\tau(a_\tau, U^y_\tau)$ is distributed according to $p(\cdot \mid a_\tau)$. Given a realized set of seeds $U = ((U^a_{j,\tau})_{j\in I}, U^y_\tau)_{\tau\geq 1}$, define action and signal realizations under a hybrid strategy profile as follows: in each period τ, every player j prescribed to follow σ_j plays $a_{j,\tau} = f^a_{j,\tau}(h_j^{\tau-1}, U^a_{j,\tau})$, while player i plays $\widetilde{a}_i$ in the periods so prescribed, and the period-τ signal profile is $y_\tau = f^y_\tau(a_\tau, U^y_\tau)$.

Let $\overline{U} = U \setminus \{U^y_{t'}\}$, and condition on a realization of $\overline{U}$. Consider play under $\widetilde{\sigma}^{t'-1}$ and under $\widetilde{\sigma}^{t'}$, and see the following three facts. First, play through period $t'-1$ coincides under σ, $\widetilde{\sigma}^{t'-1}$, and $\widetilde{\sigma}^{t'}$; the strategies and draws of U are the exact same, meaning the action and signal realizations are the exact same. Rivals' period-t' actions are therefore the exact same, $a_{j,t'} = f^a_{j,t'}(h_j^{t'-1}, U^a_{j,t'})$, under all three of σ, $\widetilde{\sigma}^{t'-1}$, and $\widetilde{\sigma}^{t'}$, and thus the on-path $a_{t'} = (a_{i,t'}, a_{-i,t'})$ is $\overline{U}$-measurable and distributed according to $\alpha_{t'}$ under $\widetilde{\sigma}^{t'}$, while under $\widetilde{\sigma}^{t'-1}$, $a_{-i,t'}$ is distributed according to $\alpha_{-i,t'}$ as well. Second, under both $\widetilde{\sigma}^{t'-1}$ and $\widetilde{\sigma}^{t'}$, h^t_{-i} is a deterministic function of $\left(\overline{U}, (y^\ell_{-i,t'})_{\ell=0}^{\min\{t-t',L\}}\right)$, where the latter term is the components of the period-t' signal that rivals observe by period t. To see this, note that the two strategy profiles are identical after period t', and conditional on $(y^\ell_{-i,t'})_{\ell=0}^{\min\{t-t',L\}}$ and $\overline{U}$ the actions in period $t'+1$ are the same, and therefore the signals in period $t'+1$ are the same; continuing by induction through period t yields the claim.[19] Third, because $U^y_{t'}$ is uniform and independent of $\overline{U}$, the conditional distribution of $(y^\ell_{-i,t'})_{\ell=0}^{\min\{t-t',L\}}$ given $\overline{U}$ is $p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid a_{t'})$ under $\widetilde{\sigma}^{t'}$ and $p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid \widetilde{a}_i, a_{-i,t'})$ under $\widetilde{\sigma}^{t'-1}$.

Total variation distance is convex, and therefore Jensen's inequality implies that the total variation distance is at most the expected total variation distance between two conditional distributions, where the expectation is taken over the common conditioning variable. We therefore have

$$\begin{aligned}
\left\|\pi^{t'-1}_{-i} - \pi^{t'}_{-i}\right\|_{TV} &\le \mathbb{E}_{\overline{U}}\left[\left\|\pi^{t'-1}_{-i}(\cdot \mid \overline{U}) - \pi^{t'}_{-i}(\cdot \mid \overline{U})\right\|_{TV}\right] \\
&\le \mathbb{E}_{\overline{U}}\left[\left\|p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid \widetilde{a}_i, a_{-i,t'}) - p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid a_{t'})\right\|_{TV}\right] \\
&= \mathbb{E}_{a\sim\alpha_{t'}}\left[\left\|p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid \widetilde{a}_i, a_{-i}) - p_{-i}^{\le\min\{t-t',L\}}(\cdot \mid a)\right\|_{TV}\right] \\
&= \mathbb{E}_{a\sim\alpha_{t'}}\left[TV_i^{\le\min\{t-t',L\}}(a, \widetilde{a}_i)\right] = TV_i^{\le\min\{t-t',L\}}(\widetilde{a}_i, \alpha_{t'}),
\end{aligned}$$

where the second inequality is by the second and third facts—pushing the conditional distributions of $y_{-i}^{\le\min\{t-t',L\}}$ through a common deterministic function weakly decreases total variation distance by the data-processing inequality—and the first equality is by the first fact. Substituting into (40) completes the proof. □

Fix a Nash equilibrium σ of Γ, i, and $\widetilde{a}_i$. Following AK19, let $\overline{\sigma}_{-i}$ denote the *nonresponsive*

[19] Note that the total variation distance between $\pi^{t'-1}_{-i}$ and $\pi^{t'}_{-i}$—both distributions over h^t_{-i}—does not depend on $y_{i,t'}$ because under both strategies player i plays $\widetilde{a}_i$ regardless of histories, and does not depend on $y^\ell_{-i,t'}$ for $\ell > t-t'$ because such signals are not observed by the end of period t.

strategy for players $-i$ that plays the on-path marginal $\alpha_{-i,t} \in \Delta(A_{-i})$ in period t after every history, and decompose the value of the permanent deviation as (Awaya and Krishna, 2019, equation (5))

$$v_i(\widetilde{\sigma}_i, \sigma_{-i}) - v_i(\sigma) = \underbrace{v_i(\widetilde{\sigma}_i, \sigma_{-i}) - v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i})}_{\text{loss from punishment}} + \underbrace{v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i}) - v_i(\sigma)}_{\text{gain when unpunished}}, \tag{41}$$

where $v_i(\cdot)$ denotes player i's normalized discounted expected payoff. Because σ is a Nash equilibrium, the left-hand side is nonpositive.

The gain when unpunished is the static gain. Against the nonresponsive strategy, player i's expected flow payoff in period t is $u_i(\widetilde{a}_i, \alpha_{-i,t})$, while its equilibrium expected flow payoff is $u_i(\alpha_t)$. Because both are linear in the period-t distribution of play,

$$v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i}) - v_i(\sigma) = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1}\left(u_i(\widetilde{a}_i, \alpha_{-i,t}) - u_i(\alpha_t)\right) = u_i(\widetilde{a}_i, \alpha_{-i}) - u_i(\alpha) = g_i(\widetilde{a}_i, \alpha),$$

where α_{-i} is the marginal of the occupation measure α on A_{-i}. (AK19's Lemma 4.2 establishes the corresponding claim for an arbitrary distribution attaining the equilibrium payoff via a minimax and convexity argument; because (39) is stated at the occupation measure itself, linearity suffices.)

The loss from punishment is controlled by history divergence. Player i plays $\widetilde{a}_i$ in every period in both terms of the loss from punishment, so the two payoffs differ only through the distribution of rivals' play. Let $\phi_t(h_{-i}^{t-1}) = \sum_{a_{-i}} \sigma_{-i}(a_{-i} \mid h_{-i}^{t-1})\, u_i(\widetilde{a}_i, a_{-i})$ be player i's expected period-t flow payoff given rivals' history, and note that $\|\phi_t\|_\infty \le \|u_i\|_\infty$. Under $(\widetilde{\sigma}_i, \sigma_{-i})$, rivals' period-$t$ histories are distributed according to $\widetilde{\mu}_{-i}^{t-1}$, while the nonresponsive strategy replicates rivals' on-path behavior, $\alpha_{-i,t}(a_{-i}) = \sum_{h_{-i}^{t-1}} \mu_{-i}^{t-1}(h_{-i}^{t-1})\, \sigma_{-i}(a_{-i} \mid h_{-i}^{t-1})$. Therefore

$$\begin{aligned}|v_i(\widetilde{\sigma}_i, \sigma_{-i}) - v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i})| &= \left|(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\sum_{h_{-i}^{t-1}}\left(\widetilde{\mu}_{-i}^{t-1}(h_{-i}^{t-1}) - \mu_{-i}^{t-1}(h_{-i}^{t-1})\right)\phi_t(h_{-i}^{t-1})\right| \\ &\le 2\|u_i\|_\infty\,(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\left\|\widetilde{\mu}_{-i}^{t-1} - \mu_{-i}^{t-1}\right\|_{TV},\end{aligned}$$

using $|\sum_h [\widetilde{\mu}(h) - \mu(h)]\phi(h)| \le 2\|\phi\|_\infty \|\widetilde{\mu} - \mu\|_{TV}$.

Now apply Lemma 5 at horizon $t-1$ and exchange the order of summation, substituting

$s = t - 1 - t'$:

$$\sum_{t=1}^{\infty} \delta^{t-1} \left\| \widetilde{\mu}_{-i}^{t-1} - \mu_{-i}^{t-1} \right\|_{TV} \le \sum_{t=2}^{\infty} \delta^{t-1} \sum_{t'=1}^{t-1} TV_i^{\le \min\{t-1-t',\, L\}}(\widetilde{a}_i, \alpha_{t'})$$
$$= \sum_{t'=1}^{\infty} \sum_{s=0}^{\infty} \delta^{s+t'}\, TV_i^{\le \min\{s,\, L\}}(\widetilde{a}_i, \alpha_{t'})$$
$$= \sum_{t'=1}^{\infty} \delta^{t'} \left[\sum_{s=0}^{L-1} \delta^s\, TV_i^{\le s}(\widetilde{a}_i, \alpha_{t'}) + \frac{\delta^L}{1-\delta}\, TV_i^{\le L}(\widetilde{a}_i, \alpha_{t'}) \right].$$

Because each $TV_i^{\le s}(\widetilde{a}_i, \cdot)$ is linear in the distribution and $(1-\delta) \sum_{t'} \delta^{t'-1} \alpha_{t'} = \alpha$, multiplying by $(1-\delta)$ gives

$$(1-\delta) \sum_{t=1}^{\infty} \delta^{t-1} \left\| \widetilde{\mu}_{-i}^{t-1} - \mu_{-i}^{t-1} \right\|_{TV} \le \delta \left[\sum_{s=0}^{L-1} \delta^s\, TV_i^{\le s}(\widetilde{a}_i, \alpha) + \frac{\delta^L}{1-\delta}\, TV_i^{\le L}(\widetilde{a}_i, \alpha) \right]$$
$$= \frac{\delta}{1-\delta}\, TV_{i,\mathrm{eff}}(\widetilde{a}_i, \alpha),$$

the last equality by (38). The loss from punishment is therefore at most $\frac{2\delta}{1-\delta} \|u_i\|_\infty\, TV_{i,\mathrm{eff}}(\widetilde{a}_i, \alpha)$ in absolute value. Combining with (41) and the nonpositivity of its left-hand side,

$$g_i(\widetilde{a}_i, \alpha) = v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i}) - v_i(\sigma) \le -\left(v_i(\widetilde{\sigma}_i, \sigma_{-i}) - v_i(\widetilde{\sigma}_i, \overline{\sigma}_{-i}) \right) \le \frac{2\delta}{1-\delta}\, \|u_i\|_\infty\, TV_{i,\mathrm{eff}}(\widetilde{a}_i, \alpha),$$

which is (39). Maximizing the right-hand side over i and $\widetilde{a}_i$ delivers the second claim. □

A.4 Proof of Proposition 3

We first establish the following Lemma.

Lemma 6. *Fix α, i, and $\widetilde{a}_i$. Suppose that the conditions of Proposition 5(ii)—the delayed-signal counterpart of Proposition 1(ii)—hold, and*

$$V_i(\alpha) \le \frac{\delta}{1-\delta}\, p_{\min}\, \|u_i\|_\infty^2 \sum_a \alpha(a)\, TV_{i,\mathrm{eff}}(a, \widetilde{a}_i). \tag{42}$$

If α satisfies (bound) at the constant manipulation $\widetilde{a}_i$, then it satisfies (bound-AK19) at $\widetilde{a}_i$.

Proof. Throughout, fix i and $\widetilde{a}_i$, let $\overline{s}_i$ be the constant manipulation with $\overline{s}_i(\widetilde{a}_i \mid a_i) = 1$ for all a_i, and write $TV_i(\alpha) = \sum_a \alpha(a) TV_{i,\mathrm{eff}}(a, \widetilde{a}_i)$ and $\chi_i^2(\alpha) = \chi_{i,\mathrm{eff}}^2(\overline{s}_i, \alpha; p_{-i})$. Note that because $p_{-i}^{\le \ell}(y_{-i}^{\le \ell} \mid a)$ is a sum of probabilities over full signal profiles, it is at least $p_{\min}$, and hence $\chi_i^2(\widetilde{a}_i, a; p_{-i}^{\le \ell}) \le \frac{4}{p_{\min}} TV_i^{\le \ell}(a, \widetilde{a}_i)^2$. Because total variation is at most one,

$TV_i^{\le \ell}(a,\widetilde{a}_i)^2 \le TV_i^{\le \ell}(a,\widetilde{a}_i)$. Combining these inequalities, weighting by $w_\ell = (1-\delta)\delta^\ell$ for $\ell < L$ and δ^L for $\ell = L$, and averaging over α yields

$$\begin{aligned}
\chi_i^2(\alpha) &\le \frac{4}{p_{\min}} \sum_a \alpha(a) \sum_{\ell=0}^{L} w_\ell TV_i^{\le \ell}(a,\widetilde{a}_i)^2 \\
&\le \frac{4}{p_{\min}} \sum_a \alpha(a) \sum_{\ell=0}^{L} w_\ell TV_i^{\le \ell}(a,\widetilde{a}_i) \\
&= \frac{4}{p_{\min}} TV_i(\alpha).
\end{aligned} \tag{43}$$

By hypothesis, Proposition 5(ii) is applicable, so the right-hand side of (bound) at $\overline{s}_i$ is $\sqrt{\frac{\delta}{1-\delta}\chi_i^2(\alpha)V_i(\alpha)}$. Suppose α satisfies (bound) at $\overline{s}_i$. Combining (43) with (42),

$$\begin{aligned}
g_i(\widetilde{a}_i,\alpha) &\le \sqrt{\frac{\delta}{1-\delta}\chi_i^2(\alpha)V_i(\alpha)} \\
&\le \sqrt{\frac{\delta}{1-\delta}\cdot\frac{4TV_i(\alpha)}{p_{\min}}\cdot\frac{\delta}{1-\delta}p_{\min}\|u_i\|_\infty^2\, TV_i(\alpha)} \\
&= \frac{2\delta}{1-\delta}\|u_i\|_\infty\, TV_i(\alpha),
\end{aligned}$$

which is (bound-AK19). □

Because every term of (38) is nonnegative, $TV_{i,\mathrm{eff}}(a,\widetilde{a}_i) \ge \delta^L TV_i^{\le L}(a,\widetilde{a}_i)$, so it suffices to show that $V_i(\alpha) \le \frac{\delta^{L+1}}{1-\delta}p_{\min}\|u_i\|_\infty^2 \sum_a \alpha(a)TV_i^{\le L}(a,\widetilde{a}_i)$ for every α, i, and $\widetilde{a}_i$ once δ is large enough. Let

$$M_i = \max_{\widetilde{a}_i\in A_i}\max_{a,a'\in A}\frac{\left(u_i(a)-u_i(a')\right)^2}{TV_i^{\le L}(a,\widetilde{a}_i)+TV_i^{\le L}(a',\widetilde{a}_i)},$$

with the convention $0/0 = 0$. Each M_i is finite because A is finite and, by hypothesis, the denominator is only zero whenever the numerator is. Further, notice that if a and a' are both independently and identically distributed according to α, we can rewrite $V_i(\alpha) = \frac{1}{2}\mathbb{E}[(u_i(a)-u_i(a'))^2]$. Relating this to M_i yields

$$\begin{aligned}
V_i(\alpha) &= \frac{1}{2}\sum_{a,a'}\alpha(a)\alpha(a')\left(u_i(a)-u_i(a')\right)^2 \\
&\le \frac{M_i}{2}\sum_{a,a'}\alpha(a)\alpha(a')\left(TV_i^{\le L}(a,\widetilde{a}_i)+TV_i^{\le L}(a',\widetilde{a}_i)\right)
\end{aligned}$$

$$= M_i \sum_a \alpha(a) TV_i^{\leq L}(a, \widetilde{a}_i).$$

Therefore, (42) holds for every α and $\widetilde{a}_i$ whenever $\frac{\delta^{L+1}}{1-\delta} p_{\min} \|u_i\|_\infty^2 \geq M_i$ for every i. If $\|u_i\|_\infty = 0$, then $M_i = 0$ and this inequality holds for every δ; otherwise, its left-hand side is strictly increasing in δ and diverges to $+\infty$ as $\delta \to 1$. Thus there exists a threshold $\overline{\delta} < 1$ such that the inequality holds for every i if $\delta \geq \overline{\delta}$. The conclusion then follows from Lemma 6. $\square$

A.5 Proof of Lemma 4

First define the constants mentioned in the lemma statement:

$$\begin{aligned}
\kappa &= \frac{b_2^2(N-1)}{1+\rho(N-2)}, \\
\Delta &= b_1 + \frac{\rho b_2(N-1)}{1+\rho(N-2)}, \\
\sigma_{i|-i}^2 &= \frac{(1-\rho)(1+\rho(N-1))}{1+\rho(N-2)}, \\
\rho_z &= \frac{N^2\rho\sigma_z^2 - (1-\rho)(1+\rho(N-1))}{N^2\sigma_z^2 + (N-1)(1-\rho)(1+\rho(N-1))}, \\
\sigma_{i|z}^2 &= \frac{N^2\sigma_z^2 + (N-1)(1-\rho)(1+\rho(N-1))}{N^2\sigma_z^2 + N(1+\rho(N-1))}, \\
\kappa_z &= \frac{b_2^2(N-1)}{\sigma_{i|z}^2(1+\rho_z(N-2))}, \\
\Delta_z &= b_1 + \frac{\rho_z b_2(N-1)}{1+\rho_z(N-2)}, \\
\sigma_{i|-i,z}^2 &= \frac{\sigma_{i|z}^2(1-\rho_z)(1+\rho_z(N-1))}{1+\rho_z(N-2)}
\end{aligned}$$

where ρ_z and $\sigma_{i|z}^2$ are the correlation and variance of firms' sales conditional on z respectively. Suppose that rivals set prices p_{-i}, and firm i is recommended to set p_i but instead sets $\widetilde{p}_i$. Let $\{f(\cdot \mid p)\}_{p\in[c,\overline{p}]^N}$ be the family of conditional distributions of log sales given prices. Under p and $(\widetilde{p}_i, p_{-i})$, rivals' sales are distributed according to $\mathcal{N}(\mu_{-i}(p), \Sigma_{-i})$ and $\mathcal{N}(\mu_{-i}(\widetilde{p}_i, p_{-i}), \Sigma_{-i})$ respectively, where $\mu_{-i}(\cdot)$ and Σ_{-i} are the subvector and submatrix of $\mu(\cdot)$ and Σ eliminating firm i's entries, $\mu_{-i}(p) - \mu_{-i}(\widetilde{p}_i, p_{-i}) = b_2(p_i - \widetilde{p}_i)\mathbf{1}_{N-1}$, and $\mathbf{1}_K$ is a K-dimensional vector of ones.

Using the formula for χ^2-divergence of two normally distributed vectors with the same covariance, the detectability from rivals' sales only can be written as

$$\begin{aligned} 1+\chi_i^2(s_i,\alpha;f_{-i}) &= \mathbb{E}\left[\exp\left(\left(b_2(p_i-\widetilde{p}_i)\right)^2 \mathbf{1}'\Sigma_{-i}^{-1}\mathbf{1}\right)\right] \\ &= \mathbb{E}\left[\exp\left(\left(p_i-\widetilde{p}_i\right)^2\kappa\right)\right]. \end{aligned} \tag{44}$$

Meanwhile, $x_i \mid x_{-i}, p$ is also normally distributed with mean given by

$$\mu_{i|-i}(p) = A - b_1 p_i + b_2 \sum_{j\neq i} p_j + \frac{\rho}{1+\rho(N-2)} \sum_{j\neq i} \left(x_j - A + b_1 p_j - b_2 \sum_{k\neq j} p_k \right)$$

and variance $\sigma^2_{i|-i}$. Firm i setting $\widetilde{p}_i$ instead of p_i shifts this conditional mean by $\Delta(p_i-\widetilde{p}_i)$ and leaves $\sigma^2_{i|-i}$ unchanged. Because the shift is constant in x_{-i}, the f-divergence chain rule factorizes the per-deviation detectability of the full signal (x_i, x_{-i}) into the rivals-only term (44) and the detectability of x_i conditional on x_{-i}:[20] for each $(p,\widetilde{p}_i)$,

$$1+\chi^{2,T}(p,\widetilde{p}_i) = \exp\left((p_i-\widetilde{p}_i)^2\kappa\right)\exp\left(\frac{\left(\Delta(p_i-\widetilde{p}_i)\right)^2}{\sigma^2_{i|-i}}\right), \tag{45}$$

the second factor being one plus the χ^2-divergence between two normals of common variance $\sigma^2_{i|-i}$ whose means differ by $\Delta(p_i-\widetilde{p}_i)$. Averaging (45) over $(p,\widetilde{p}_i)\sim(\alpha,s_i)$ yields the case (b) detectability, $1+\chi_i^{2,T}(s_i,\alpha) = \mathbb{E}\left[\exp\left((p_i-\widetilde{p}_i)^2\left(\kappa+\frac{\Delta^2}{\sigma^2_{i|-i}}\right)\right)\right]$. Because case (b) features i-semipublic monitoring, Corollary 1 gives $\underline{\chi}^{2,S}_{i,\text{eff}}(s_i,\alpha) = \chi^{2,T}_{i,\text{eff}}(s_i,\alpha)$, which is this expression.

In case (a), the mediator observes only x_{-i} and i's report. The deterministic manipulation reporting $\widetilde{x}_i = x_i - \Delta(p_i-\widetilde{p}_i)$ cancels the conditional mean shift, so the second factor of (45) becomes one. The chain rule then leaves only the rivals-only term, $1+\underline{\chi}^{2,NS}_{i,\text{eff}}(s_i,\alpha) = \mathbb{E}\left[\exp\left((p_i-\widetilde{p}_i)^2\kappa\right)\right]$, which is optimal by Proposition 1.[21] Because $\Delta>0$ (the map $\rho\mapsto\rho(N-1)/(1+\rho(N-2))$ is increasing on $(-1/(N-1),1)$, so $\Delta\geq b_1-(N-1)b_2=\beta>0$ by Assumption 1(i); the same argument at ρ_z gives $\Delta_z\geq\beta>0$) and by hypothesis there is positive probability of $p_i\neq\widetilde{p}_i$, the term inside the second exponential is strictly positive with positive probability and $\underline{\chi}^{2,NS}_{i,\text{eff}} < \chi^{2,S}_{i,\text{eff}}$.

For case (c), notice that $x\mid p,z\sim\mathcal{N}(\mu(p,z),\Sigma_z)$, where $\mu_i(p,z) = A-b_1p_i+b_2\sum_{j\neq i}p_j + \frac{1+(N-1)\rho}{1+(N-1)\rho+N\sigma_z^2}z$ and Σ_z has $\sigma^2_{i|z}$ on the diagonals and $\rho_z\sigma^2_{i|z}$ on the off-diagonals. Firms observe

[20] This is the multiplicative form of the decomposition in Lemma 3, available because the conditional detectability is constant in x_{-i}.

[21] Another way of seeing this is to note that case (a) falls under Corollary 1(ii), with $g_i(p_i) = -\left(b_1+\frac{b_2\rho(N-1)}{1+\rho(N-2)}\right)p_i$, $g_{-i}(p_{-i}) = \frac{A(1-\rho)}{1+\rho(N-2)} + \frac{\rho b_1+b_2}{1+\rho(N-2)}\sum_{j\neq i}p_j$, and the residual equal to $x_i-\mu_{i|-i}(p)+\frac{\rho}{1+\rho(N-2)}\sum_{j\neq i}x_j$. The residual is independent of the action profile conditional on x_{-i}, as $x_i-\mu_{i|-i}(p)\sim\mathcal{N}(0,\sigma^2_{i|-i})$ and the remaining term is conditionally deterministic.

sales contemporaneously and z after L periods, so detection uses the information set $(\widetilde{x}_i, x_{-i})$ at lags $\ell < L$ and (x_{-i}, z) at lag L. Because z is public, any message manipulation that places probability less than one on truthful reporting of z results in an effective detectability of ∞, and hence we only study the choice over message manipulations of own sales $q_i(\widetilde{x}_i \mid p_i, \widetilde{p}_i, x_i)$ while implicitly fixing $q_i^L(z \mid p_i, \widetilde{p}_i, x_i, \widetilde{x}_i, z) = 1$.

Similar to cases (a) and (b), $x_i \mid x_{-i}, z, p$ is normal, with variance $\sigma^2_{i|-i,z}$ and a mean that the deviation shifts by $\Delta_z(p_i - \widetilde{p}_i)$. Suppose that the manipulation takes the form of a constant shift $q_i(x_i - \Delta^*(p_i, \widetilde{p}_i)(p_i - \widetilde{p}_i) \mid p_i, \widetilde{p}_i, x_i) = 1$ for some per-unit recentering $\Delta^*(p_i, \widetilde{p}_i)$. Then, applying the chain rule of (45) at each information set—before z the residual conditional mean shift is $(\Delta - \Delta^*)(p_i - \widetilde{p}_i)$, and after z it is $(\Delta_z - \Delta^*)(p_i - \widetilde{p}_i)$—and noting that z carries no detectability of its own, the effective detectability takes the form of (32). Among such manipulations, the minimizer $\Delta^*(p_i, \widetilde{p}_i)$ of (32) lies between Δ_z and Δ for every $(p_i, \widetilde{p}_i)$ (strictly so when $L > 0$, as the derivative of (32) is nonzero at each endpoint). If $L = 0$, the first term has weight zero and the minimizer is $\Delta^* = \Delta_z$, so (32) reduces to $\mathbb{E}\left[\exp\left((p_i - \widetilde{p}_i)^2 \kappa_z\right)\right]$, the rivals-only detectability conditional on z. If $L > 0$, minimizing the first term alone requires $\Delta^* = \Delta$ while minimizing the second alone requires $\Delta^* = \Delta_z$. Since $\Delta_z \neq \Delta$, no single Δ^* does both, so at least one term has a strictly positive residual and detectability is strictly above the rivals-only floor. Meanwhile, truthful reporting corresponds to the recentering $0 < \Delta_z$, and both terms of (32) are strictly decreasing in the recentering on $(-\infty, \Delta_z]$, which gives the strict inequality $\underline{\chi}^{2,z(L)}_{i,\text{eff}} < \chi^{2,T}_{i,\text{eff}}$ in the statement.

The final step is to show that such a constant-shift manipulation is globally optimal. By Proposition 6, it is if and only if, for every received x_i, the effective excess score function $\Lambda(\widetilde{x}_i \mid x_i)$ is minimized over $\widetilde{x}_i$ at $x_i - s^*$, with $s^* \equiv \Delta^*(p_i, \widetilde{p}_i)(p_i - \widetilde{p}_i)$. To facilitate more readable notation, let $\ell \in \{0, L\}$ denote the lag and $\mathcal{I}_\ell$ denote conditioned-on information at lag ℓ with $\mathcal{I}_0 = x_{-i}$ and $\mathcal{I}_L = (x_{-i}, z)$. The excess score function is

$$
\begin{aligned}
\Lambda(\widetilde{x}_i \mid x_i) &= (1 - \delta^L)\Lambda_0(\widetilde{x}_i \mid x_i) + \delta^L \Lambda_L(\widetilde{x}_i \mid x_i), \\
\Lambda_\ell(\widetilde{x}_i \mid x_i) &= \mathbb{E}_{p_{-i}}\left[\int \frac{f(\mathcal{I}_\ell \mid \widetilde{p}_i)^2}{f(\mathcal{I}_\ell \mid p_i)} f(x_i \mid \widetilde{p}_i, \mathcal{I}_\ell) \left(\frac{f(\widetilde{x}_i + s^* \mid \widetilde{p}_i, \mathcal{I}_\ell) - f(\widetilde{x}_i \mid p_i, \mathcal{I}_\ell)}{f(\widetilde{x}_i \mid p_i, \mathcal{I}_\ell)}\right) d\mathcal{I}_\ell\right],
\end{aligned}
$$

where all densities inside the expectation also condition on p_{-i}, $\mathbb{E}_{p_{-i}}$ averages over $p_{-i} \sim \alpha(\cdot \mid p_i)$, and the inner integral runs over $\mathcal{I}_\ell$ (for $\ell = L$, over (x_{-i}, z) with z drawn from its marginal).[22] Let $Z_\ell = \int f(\mathcal{I}_\ell \mid \widetilde{p}_i, p_{-i})^2 / f(\mathcal{I}_\ell \mid p_i, p_{-i}) d\mathcal{I}_\ell$, and $\widehat{f}_\ell(\mathcal{I}_\ell \mid p_{-i}) = Z_\ell^{-1} f(\mathcal{I}_\ell \mid$

[22] Λ_L averages over z because $\widetilde{x}_i$ is sent before z is observed, so the manipulation $q_i(\widetilde{x}_i \mid p_i, \widetilde{p}_i, x_i)$ cannot condition on z. Technically, this statement is only true for $L > 0$—if $L = 0$ then $\widetilde{x}_i$ can condition on z. However, for the $L = 0$ case, Δ^* not depending on the realization of z turns out to attain the lower bound and is thus optimal, so not conditioning on z is without loss for this case.

$\widetilde{p}_i, p_{-i})^2/f(\mathcal{I}_\ell \mid p_i, p_{-i})$, so that $\widehat{f}_\ell(\cdot \mid p_{-i})$ is a density. We can rewrite Λ_ℓ as

$$\Lambda_\ell(\widetilde{x}_i \mid x_i) = \mathbb{E}_{p_{-i}}\left[Z_\ell \mathbb{E}_{\mathcal{I}_\ell}\left[f(x_i \mid \widetilde{p}_i, p_{-i}, \mathcal{I}_\ell)\left(\frac{f(\widetilde{x}_i + s^* \mid \widetilde{p}_i, p_{-i}, \mathcal{I}_\ell) - f(\widetilde{x}_i \mid p_i, p_{-i}, \mathcal{I}_\ell)}{f(\widetilde{x}_i \mid p_i, p_{-i}, \mathcal{I}_\ell)}\right)\right]\right]$$

where $\mathbb{E}_{\mathcal{I}_\ell}$ is taken with respect to $\widehat{f}_\ell(\cdot \mid p_{-i})$.

Fix $p = (p_i, p_{-i})$ and analyze the inner expectation. Let $\mu_\ell(\mathcal{I}_\ell) \equiv \mathbb{E}[x_i \mid \mathcal{I}_\ell, p]$, which is affine in $\mathcal{I}_\ell$, and abbreviate $\sigma_0^2 = \sigma_{i|-i}^2$, $\sigma_L^2 = \sigma_{i|-i,z}^2$, $\Delta_0 = \Delta$, and $\Delta_L = \Delta_z$. Conditional on $p_{-i}, \mathcal{I}_\ell$, x_i is distributed according to $\mathcal{N}(\mu_\ell(\mathcal{I}_\ell), \sigma_\ell^2)$ on path and $\mathcal{N}(\mu_\ell(\mathcal{I}_\ell) + s_\ell, \sigma_\ell^2)$ under the deviation, where $s_\ell \equiv \Delta_\ell(p_i - \widetilde{p}_i)$. Thus, letting $\phi(\cdot; \mu, \sigma^2)$ be the $\mathcal{N}(\mu, \sigma^2)$ density function and $\Delta s_\ell \equiv s^* - s_\ell$, and noting that $\phi(z; \mu_1, \sigma^2)/\phi(z; \mu_2, \sigma^2) = \exp\left(\frac{z}{\sigma^2}(\mu_1 - \mu_2) - \frac{1}{2\sigma^2}(\mu_1^2 - \mu_2^2)\right)$, we can rewrite $\Lambda_\ell(\widetilde{x}_i \mid x_i) = \mathbb{E}_{p_{-i}}\left[Z_\ell \mathbb{E}_{\mathcal{I}_\ell}\left[g(\mu_\ell(\mathcal{I}_\ell))\right]\right]$, where

$$g(\mu_\ell) \equiv \phi(x_i; \mu_\ell(\mathcal{I}_\ell) + s_\ell, \sigma_\ell^2)\left(\exp\left(-\frac{\Delta s_\ell(\widetilde{x}_i - \mu_\ell(\mathcal{I}_\ell))}{\sigma_\ell^2} - \frac{\Delta s_\ell^2}{2\sigma_\ell^2}\right) - 1\right). \tag{46}$$

To evaluate $\mathbb{E}_\ell$, we use the following properties of $\widehat{f}_\ell$. First, because $\mathcal{I}_\ell \mid p$ and $\mathcal{I}_\ell \mid \widetilde{p}_i, p_{-i}$ are both Gaussian with a common covariance, $\widehat{f}_\ell \propto f(\cdot \mid \widetilde{p}_i, \dots)^2/f(\cdot \mid p_i, \dots)$, $\mathcal{I}_\ell \sim \widehat{f}_\ell$ is also Gaussian with the (i) same covariance and (ii) a mean shifted by twice the deviation's shift of the on-path mean.[23] Since $\mu_\ell(\mathcal{I}_\ell)$ is an affine function, it follows that $\mu_\ell(\mathcal{I}_\ell)$ is also normally distributed, and we can write $\mu_\ell(\mathcal{I}_\ell) \sim \mathcal{N}(m_\ell, v_\ell)$. Due to (i) and the law of total variance, $v_\ell + \sigma_\ell^2 = \mathrm{Var}(x_i \mid p) = 1$, and (ii) implies that $m_\ell = \mu_i(p) + 2(b_1 - \Delta_\ell)(p_i - \widetilde{p}_i)$. Because $g(\cdot)$ only depends on $\mathcal{I}_\ell$ through μ_ℓ, we may therefore rewrite $\mathbb{E}_{\mathcal{I}_\ell}[g(\mu_\ell(\mathcal{I}_\ell))]$ as $\mathbb{E}_{\mu_\ell \sim \mathcal{N}(m_\ell, v_\ell)}[g(\mu_\ell)]$. Direct computation, noting that $\mathbb{E}_{\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)}[\phi(x; \mu, \sigma^2)] = \phi(x; \mu_0, \sigma_0^2 + \sigma^2)$, yields

$$\begin{aligned}
\mathbb{E}_{\mu_\ell \sim \mathcal{N}(m_\ell, v_\ell)}[g(\mu_\ell)] &= \gamma_\ell \phi\left(x_i; k_\ell, 1\right) \exp\left(-\frac{\Delta s_\ell}{\sigma_\ell^2}\widetilde{x}_i\right) - \phi(x_i; \hat{k}_\ell, 1), \\
k_\ell &\equiv \mu_i(p) + (2b_1 - \Delta_\ell)(p_i - \widetilde{p}_i) + \frac{\Delta s_\ell}{\sigma_\ell^2}(1 - \sigma_\ell^2), \\
\hat{k}_\ell &\equiv k_\ell - \frac{\Delta s_\ell}{\sigma_\ell^2}(1 - \sigma_\ell^2), \\
\gamma_\ell &\equiv \exp\left(\frac{\Delta s_\ell(\mu_i(p) + 2(b_1 - \Delta_\ell)(p_i - \widetilde{p}_i))}{\sigma_\ell^2} + \frac{\Delta s_\ell^2(1 - 2\sigma_\ell^2)}{2\sigma_\ell^4}\right)
\end{aligned}$$

Removing terms that do not depend on $\widetilde{x}_i$, we thus ultimately need to show that, for all x_i,

[23] To see this, note that $\phi(x; \mu - d, \Sigma)^2/\phi(x; \mu, \Sigma) = k^2/k \exp(-\frac{1}{2}(2(x - \mu + d)'\Sigma^{-1}(x - \mu + d) - (x - \mu)'\Sigma^{-1}(x - \mu))) = k \exp(-\frac{1}{2}(x - \mu + 2d)'\Sigma^{-1}(x - \mu + 2d) + d'\Sigma^{-1}d) = \exp(d'\Sigma^{-1}d)\phi(x; \mu - 2d, \Sigma)$, where $k = (2\pi)^{-\dim x/2}|\Sigma|^{-1/2}$.

$$x_i - s^* \in \arg\min_{\widetilde{x}_i} \mathbb{E}_{p_{-i}}\Bigg[(1-\delta^L) Z_0 \gamma_0 \phi(x_i; k_0, 1) \exp\left(-\frac{\Delta s_0}{\sigma_0^2}\widetilde{x}_i\right) + \delta^L Z_L \gamma_L \phi(x_i; k_L, 1) \exp\left(-\frac{\Delta s_L}{\sigma_L^2}\widetilde{x}_i\right)\Bigg].$$

Consider a relaxed problem in which $\widetilde{x}_i$ is allowed to condition on p_{-i} as well. Clearly if $x_i - s^*$ is a solution to the relaxed problem point-by-point, where s^* does not depend on p_{-i}, it is a solution when averaging over p_{-i} as well. The objective of the relaxed problem is convex, being a positive combination of exponentials in $\widetilde{x}_i$; as $\Delta^* \in (\Delta_z, \Delta)$, Δs_0 and Δs_L have opposite signs, an interior minimum exists, and the first-order condition is necessary and sufficient for a global minimum. Its first-order condition, after taking logs and noting that the log-ratio of two unit-variance Gaussian PDFs is linear in x_i with slope equal to the difference in means, is

$$\begin{aligned}
\widetilde{x}_i^*(x_i) &= \frac{k_L - k_0}{\Delta s_L/\sigma_L^2 - \Delta s_0/\sigma_0^2} x_i - \overline{s}, \\
\overline{s} &= \left(\frac{\Delta s_0}{\sigma_0^2} - \frac{\Delta s_L}{\sigma_L^2}\right)^{-1} \left[\log\left(-\frac{\Delta s_L/\sigma_L^2}{\Delta s_0/\sigma_0^2} \cdot \frac{\delta^L}{1-\delta^L} \cdot \frac{Z_L \gamma_L}{Z_0 \gamma_0}\right) - \frac{1}{2}\left(k_L^2 - k_0^2\right)\right].
\end{aligned}$$

The coefficient on x_i evaluates to 1, as

$$\begin{aligned}
k_L - k_0 &= (\Delta_0 - \Delta_L)(p_i - \widetilde{p}_i) + \frac{\Delta s_L}{\sigma_L^2} - \frac{\Delta s_0}{\sigma_0^2} + \Delta s_0 - \Delta s_L \\
&= s_0 - s_L + \frac{\Delta s_L}{\sigma_L^2} - \frac{\Delta s_0}{\sigma_0^2} + s_L - s_0 \\
&= \frac{\Delta s_L}{\sigma_L^2} - \frac{\Delta s_0}{\sigma_0^2}.
\end{aligned}$$

Further, the only way in which $\overline{s}$ depends on p_{-i} is through $\mu_i(p)$ in γ_ℓ and k_ℓ. However, $\log(\gamma_L/\gamma_0) = (\Delta s_L/\sigma_L^2 - \Delta s_0/\sigma_0^2)\mu_i(p) + \overline{\gamma}$, and $(k_L^2 - k_0^2)/2 = ((\Delta_0 - \Delta_L)(p_i - \widetilde{p}_i) + \Delta s_L/\sigma_L^2 - \Delta s_0/\sigma_0^2 + \Delta s_0 - \Delta s_L)\mu_i(p) + \overline{k} = (\Delta s_L/\sigma_L^2 - \Delta s_0/\sigma_0^2)\mu_i(p) + \overline{k}$, where $\overline{\gamma}$ and $\overline{k}$ do not depend on $\mu_i(p)$. $\mu_i(p)$ therefore cancels out of $\overline{s}$. The solution to the relaxed problem is thus $\widetilde{x}_i = x_i - \overline{s}$, where $\overline{s}$ does not depend on p_{-i}; as such, this is the solution to the original problem as well.

Finally, assume towards a contradiction that $\overline{s} \neq s^*$. If so, then for every received x_i, firm i would prefer to move its report in the same direction, so effective detectability would strictly fall within the one-parameter family of constant shifts. This contradicts that $\Delta^*(p_i, \widetilde{p}_i)$ minimizes (32) within exactly that family. Thus, $\widetilde{x}_i = x_i - s^*$ minimizes the effective excess score function when it is defined according to the same manipulation which, by Proposition 6, implies it is the globally optimal manipulation. $\square$

A.6 Proof of Proposition 4

We begin by putting detectability in the three monitoring structures in a common form. Fix a monitoring structure o and a firm i. For a given $(p_i, \widetilde{p}_i)$, where $\Delta p = \widetilde{p}_i - p_i$, each firm chooses a fixed message manipulation d_i, where (as Lemma 4 showed) d_i is a single recentering constant for each Δp, to solve a problem of the form

$$\underline{\chi}^2_{i,\text{eff}}(\Delta p) = \min_{d \in \mathcal{D}} \sum_{\ell \in \mathcal{L}} w_\ell \left(\exp(K_\ell(d)^2 \Delta p^2) - 1 \right) \tag{47}$$

where $\mathcal{D}$ is a compact set of possible manipulations, $\mathcal{L} = \{0\}$ and $w_0 = 1$ in the no sharing and sharing scenarios, and $\mathcal{L} = \{0, L\}$, $w_0 = 1 - \delta^L$, and $w_L = \delta^L$ for the demand signal scenario. $K_0(d)^2 = \kappa + (\Delta - d)^2 / \sigma^2_{i|-i}$ (we take $\mathcal{D} = \{0\}$ when sharing and $\mathcal{D} = [0, \Delta]$ otherwise, which contains the relevant minimizers by Lemma 4), and $K_L(d)^2 = \kappa_z + (\Delta_z - d)^2 / \sigma^2_{i|-i,z}$ in the demand signal case. Because this formula is pointwise in $(\widetilde{p}_i, p_{-i})$, the overall minimized detectability is the $(p_i, \widetilde{p}_i) \sim (\alpha_i, s_i)$-average of the pointwise detectabilities,

$$\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha) = \mathbb{E}_{p_i \sim \alpha_i, \widetilde{p}_i \sim s_i(\cdot|p_i)} \left[\underline{\chi}^2_{i,\text{eff}}(\widetilde{p}_i - p_i) \right]. \tag{48}$$

Given (47), define the *linearized problem* for a given monitoring structure and its squared value as

$$K_o^2 = \min_d \sum_\ell w_\ell K_\ell(d)^2. \tag{49}$$

The solutions to this problem for each case are the monitoring constants defined in the proposition statement. Letting $d(\Delta p)$ be the solution to (47), note the following two facts. First, $e^z - 1 \geq z$ for $z \geq 0$ implies that $\underline{\chi}^2_{i,\text{eff}}(\Delta p) \geq \sum_\ell w_\ell K_\ell(d(\Delta p))^2 \Delta p^2 \geq K_o^2 \Delta p^2$, where the last inequality is by optimality of K_o. Second, letting $\overline{K}^2 = \max_{\ell,d} K_\ell(d)^2$, which is finite by compactness and continuity, $e^z - 1 \leq z + \frac{1}{2} z^2 e^z$ implies that

$$K_o^2 \Delta p^2 \leq \underline{\chi}^2_{i,\text{eff}}(\Delta p) \leq K_o^2 \Delta p^2 + \frac{1}{2} \overline{K}^4 \exp(\overline{K}^2 (\overline{p} - c)^2) \Delta p^4. \tag{50}$$

Necessity. Suppose α satisfies (bound) for all i, s_i. Let $\overline{D} = \sup_{p_i \in [c, \overline{p}]} |D_i(p_i; \alpha)| \geq 0$. If $\overline{D} = 0$, or $\mathbb{E}_{\alpha_i}[D_i(p_i; \alpha)^2] = 0$, then (34) holds trivially. Otherwise, consider a pure manipulation s_i^ε that, for each recommended p_i, places probability 1 on $\widetilde{p}_i = p_i + \varepsilon D_i(p_i; \alpha) / \overline{D}$ for a small $\varepsilon > 0$. This is a valid manipulation for ε small enough, as $D_i(p_i; \alpha)$ has the sign of

$1-b_1(p_i-c)$ and so the deviation always points towards p^N. Substituting $\Delta p = \varepsilon D_i(p_i;\alpha)/\overline{D}$ into (50) and dividing by ε^2 on both sides implies that

$$\begin{aligned}\left| K_o^2 \left(\frac{D_i(p_i;\alpha)}{\overline{D}}\right)^2 - \frac{\underline{\chi}^2_{i,\text{eff}}(\varepsilon D_i(p_i;\alpha)/\overline{D})}{\varepsilon^2}\right| &\le \frac{1}{2}\overline{K}^4 \exp(\overline{K}^2(\overline{p}-c)^2)\varepsilon^2 \left(\frac{D_i(p_i;\alpha)}{\overline{D}}\right)^4 \\ &\le \frac{1}{2}\overline{K}^4 \exp(\overline{K}^2(\overline{p}-c)^2)\varepsilon^2,\end{aligned}$$

where the last inequality is because $|D_i(p_i;\alpha)/\overline{D}| \le 1$. Taking limits as $\varepsilon \to 0$ implies that $\underline{\chi}^2_{i,\text{eff}}(\varepsilon D_i(p_i,\alpha)/\overline{D})/\varepsilon^2 \to K_o^2(D_i(p_i;\alpha)/\overline{D})^2$ uniformly in p_i.

Now, evaluating (bound) at s_i^ε and dividing both sides by ε gives

$$\mathbb{E}_\alpha\left[\frac{\pi_i(p_i+\varepsilon D_i(p_i;\alpha)/\overline{D}, p_{-i}) - \pi_i(p)}{\varepsilon}\right] \le \sqrt{\frac{\delta}{1-\delta}V_i(\alpha)\frac{\mathbb{E}_{\alpha_i}\left[\underline{\chi}^2_{i,\text{eff}}(\varepsilon D_i(p_i;\alpha)/\overline{D})\right]}{\varepsilon^2}}.$$

Taking $\varepsilon \to 0$, noting that π_i is continuously differentiable on $[c,\overline{p}]^N$ so that its difference quotients converge uniformly and integrals and limits can be interchanged, and applying uniform convergence of $\underline{\chi}^2_{i,\text{eff}}(\varepsilon D_i(p_i,\alpha)/\overline{D})/\varepsilon^2 \to K_o^2(D_i(p_i;\alpha)/\overline{D})^2$ yields

$$\mathbb{E}_{\alpha_i}\left[\frac{D_i(p_i;\alpha)^2}{\overline{D}}\right] \le \sqrt{\frac{\delta}{1-\delta}V_i(\alpha)K_o^2\mathbb{E}_{\alpha_i}\left[\frac{D_i(p_i;\alpha)^2}{\overline{D}^2}\right]}$$

which, after cancelling $\overline{D}$, squaring, and dividing through by $\mathbb{E}_{\alpha_i}[D_i(p_i;\alpha)^2]$, is (34).

Sufficiency. Suppose (34) holds for all i. First consider a pure manipulation, which takes the form $\widetilde{p}_i : [c,\overline{p}] \to [c,\overline{p}]$. As interim profits are concave, we have

$$\mathbb{E}_\alpha\left[\pi_i(\widetilde{p}_i(p_i), p_{-i}) - \pi_i(p) \mid p_i\right] \le |D_i(p_i;\alpha)|\left|\widetilde{p}_i(p_i) - p_i\right|. \tag{51}$$

We therefore have

$$\begin{aligned} g_i(\widetilde{p}_i,\alpha) &\le \mathbb{E}_{\alpha_i}\left[|D_i(p_i;\alpha)||\widetilde{p}_i(p_i)-p_i|\right] && \text{by (51)} \\ &\le \sqrt{\mathbb{E}_{\alpha_i}[D_i(p_i;\alpha)^2]\mathbb{E}_{\alpha_i}[(\widetilde{p}_i(p_i)-p_i)^2]} && \text{Cauchy–Schwarz} \\ &\le \sqrt{\frac{\delta}{1-\delta}K_o^2\mathbb{E}_{\alpha_i}[(\widetilde{p}_i(p_i)-p_i)^2]V_i(\alpha)} && \text{by (34)} \\ &\le \sqrt{\frac{\delta}{1-\delta}\underline{\chi}^2_{i,\text{eff}}(\widetilde{p}_i,\alpha)V_i(\alpha)} && \text{by (50), averaged over } \alpha \end{aligned}$$

which is (bound). For a mixed manipulation, which we can rewrite as a distribution s_i over pure manipulations $\widetilde{p}_i$, note that concavity of the square root and linearity of g_i in s_i implies that $g_i(s_i, \alpha) = \mathbb{E}[g_i(\widetilde{p}_i, \alpha)] \leq \mathbb{E}\left[\sqrt{\frac{\delta}{1-\delta}\underline{\chi}^2_{i,\text{eff}}(\widetilde{p}_i, \alpha)V_i(\alpha)}\right] \leq \sqrt{\frac{\delta}{1-\delta}\mathbb{E}\left[\underline{\chi}^2_{i,\text{eff}}(\widetilde{p}_i, \alpha)\right]V_i(\alpha)} = \sqrt{\frac{\delta}{1-\delta}\underline{\chi}^2_{i,\text{eff}}(s_i, \alpha)V_i(\alpha)}$. Thus, (bound) applying to all pure manipulations means it applies to all manipulations as well. □

References

Abreu, Dilip, Paul Milgrom, and David Pearce. 1991. "Information and Timing in Repeated Partnerships." *Econometrica* 59 (6): 1713–1733.

Awaya, Yu, and Vijay Krishna. 2016. "On Communication and Collusion." *American Economic Review* 106 (2): 285–315.

———. 2019. "Communication and Cooperation in Repeated Games." *Theoretical Economics* 14 (2): 513–553.

———. 2020. "Information Exchange in Cartels." *RAND Journal of Economics* 51 (2): 421–446.

Forges, Françoise. 1986. "An Approach to Communication Equilibria." *Econometrica* 54 (6): 1375–1385.

Fudenberg, Drew, Yuhta Ishii, and Scott Duke Kominers. 2014. "Delayed-Response Strategies in Repeated Games with Observation Lags." *Journal of Economic Theory* 150:487–514.

Fudenberg, Drew, David Levine, and Wolfgang Pesendorfer. 1998. "When are Nonanonymous Players Negligible?" *Journal of Economic Theory* 79 (1): 46–71.

Harrington, Joseph E. 2026. "A Critique of Recent Remedies for Third-Party Pricing Algorithms and Why the Solution Is Not Restrictions on Data Sharing." *Journal of Competition Law & Economics* 22 (1): 114–137.

Igami, Mitsuru, and Takuo Sugaya. 2022. "Measuring the Incentive to Collude: The Vitamin Cartels, 1990–99." *Review of Economic Studies* 89 (3): 1460–1494.

Kandori, Michihiro. 2002. "Introduction to Repeated Games with Private Monitoring." *Journal of Economic Theory* 102 (1): 1–15.

Kandori, Michihiro, and Hitoshi Matsushima. 1998. "Private Observation, Communication and Collusion." *Econometrica* 66 (3): 627–652.

Kreps, Avner A. 2026. "Information Sharing and Collusion: Theory and Evidence from Poultry Processing." Working paper.

Mailath, George J., and Larry Samuelson. 2006. *Repeated Games and Reputations: Long-Run Relationships.* Oxford University Press.

Myerson, Roger B. 1986. "Multistage Games with Communication." *Econometrica* 54 (2): 323–358.

Al-Najjar, Nabil I., and Rann Smorodinsky. 2000. "Provision of a public good with bounded cost." *Economics Letters* 67 (3): 297–301.

———. 2001. "Large nonanonymous repeated games." *Games and Economic Behavior* 37 (1): 26–39.

Polyanskiy, Yury, and Yihong Wu. 2025. *Information Theory: From Coding to Learning.* Cambridge: Cambridge University Press.

Sugaya, Takuo. 2022. "Folk Theorem in Repeated Games with Private Monitoring." *Review of Economic Studies* 89 (4): 2201–2256.

Sugaya, Takuo, and Alexander Wolitzky. 2018. "Bounding Payoffs in Repeated Games with Private Monitoring: n-Player Games." *Journal of Economic Theory* 175:58–87.

Sugaya, Takuo, and Alexander Wolitzky. 2023. “Monitoring versus Discounting in Repeated Games.” *Econometrica* 91 (5): 1727–1761.

US et al. v. Agri Stats, Inc. 2026. *Proposed Final Judgment.* No. 0:23-cv-03009-JRT-JFD, D. Minn. May 7, 2026.